

Dt - 21 sept. 2020

→ Economics is the study of household management.

defⁿ of Robbins's Economics :-

Robbins defines economics as the science which studies human behaviour as a relationship between ^{ends &} scarce means, which have alternative uses.

economic welfare → 1st & 2nd economic unit :-

→ producer → max^m profit / production / ~~satisfaction~~
→ consumer → max^m satisfaction.

→ Engineering Economics :- It is the application of economic theories in the field of engineering, is known as engineering economics.

18th century → M. Wellington, civil eng. in America introduced the concept of economics in the field of engineering (in Railway constr.)

→ Eugene L. Grant → developed engineering economics theory, Aptly Eugene L. Grant is known as the father of engineering economics.

→ components of E.E. :-

There are 2 components / scope of

E.E.

1) Physical environment :-

The technical knowledge about the machines or how to function the machine, machine design, constructions.

2) Economic / Financial Environment :-

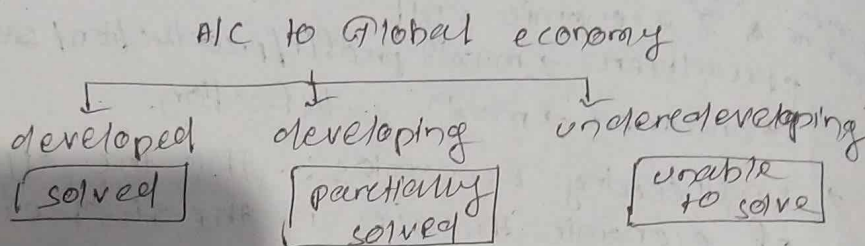
It includes the demand & supply of the product at pricing of products & factors, allocation & distribution of Resources.

The basic problem of an economy:-

There are 3 basic problems of an economy.

- 1) what to produce
- 2) how to produce
- 3) for whom to produce

1)



Inflation

persistent increase in price & demand of product

↓

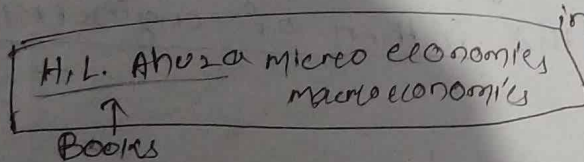
unstable growth.

Recession

persistent decrease in price & demand of product

↓

create stagnant posⁿ in economy



dt. 24/9/2020

Scope & Importance of E.E :-

- Engineering Economics play a very imp. role in all Engineering decisions.
- It is concerned with the monetary consequences and financial analysis of the projects, products & processes that engineers design.
- It helps in demand forecasting & analysis.
- It also helps in the production & cost

analysis.

- It helps
- It helps
- It also sales price
- It affects policy

Profit

Difference

Micro

* It is concerned with individual in economy

* The branch that studies the behaviour of consumer is called

* It deals with economic

* It is a rational issues.

* It covers like a product pricing, supply & demand, welfare

analysts.

- It helps in profit management.
- It helps in capital management.
- It also concerned with the product policy, sales promotion & marketing strategies.
- ~~It~~ It helps in pricing decisions, ~~policy & p~~ policies & practices.

$$\text{Profit} = \text{Revenue} - \text{cost}$$

Difference b/w micro & macro economics:-

Micro E.

* It is concerned / concentrates on all the individual unit existing in economy.

* The branch of economics that studies the behaviour of an individual consumer, ~~firm~~, family is called micro economics

* It deals with individual economics variable

* It is applied to operational or interventional issues.

* It covers various issues like demand, supply, product pricing, factor pricing, production, consumption & economic welfare etc.

Macro E.

→ It deals with aggregate variable, so it is also known as Aggregate economy.

* The branch of economics that studies the behaviour of the whole ~~an~~ economy both national & international is known as macro economics.

* It deals with aggregate economic variable.

* It is applied to environmental & external issues.

* It covers various issues like national income, general price level, distribution, employment etc.

→ As the resources available in limited number, so how to use it / utilise it efficiently, so that both producer & consumer will gain max^m satisfaction, this is the goal of Economics or subject matter.

Land
Labour
Capital
Organisation

} factors of production

* It is helpful in determining the pricing of a product along with the prices of factors of production.

* It maintains stability in general price level and resolves the major problems of the economy like inflation, deflation, unemployment, poverty & dis-equilibrium in balance of payment.

human wants are unlimited → 1st sentence of economy.

need → basic requirement

want → other than need. (things we dream of / wish)

Demand *

→ wish to get a product

→ ability to get a product

* Demands are limited in nature.

→ demand high, supply increase, price increase

→ demand less, supply decrease, price decrease.

factors

Land → get

Labour → get

Capital →

organisation

rent + salary

= GDP / GNP

↓
gross domestic product

All the inter

only trade

↓
micro

→ demand > supply

→ demand < supply

Dt:- 28 sept.

Economically
perfectly competitive

↓
price is determined

demand & supply

product

~~market~~

* when we

pay it is

eg → demand

Amount

is ready

Factors

Land $\rightarrow$ get $\rightarrow$ rent

Labour $\rightarrow$ gets $\rightarrow$ salary / wage

Capital $\rightarrow$ $n \rightarrow$ interest

Organisation $\rightarrow$ $n \rightarrow$ Profit

rent + salary / wage + interest + profit

= GDP / GNP (National Income)

gross domestic product $\rightarrow$ gross national product

All the international transactions $\rightarrow$ macro E.
only trade (supply & intake of goods)

$\downarrow$
micro Economy.

- $\rightarrow$ demand $>$ supply $\rightarrow$ price increase
- $\rightarrow$ demand $<$ supply $\rightarrow$ price decrease

Dt:- 28 sept. 2020

Economically 2 types of markets:-
perfectly competitive $\rightarrow$ imperfectly competitive

$\downarrow$
Price is decided by demand & supply of product

~~Need~~
* when want is coupled by ability to pay it is called demand.
eg $\rightarrow$ demand to get a pen.

~~Amount of product that a consumer is ready to pay for~~

Demand (Individual)

It is the amount of goods or services that a consumer is ready to take off from the market at various prevailing price during a given period of time is called demand.

Market Demand :-

It is the amount of goods & services demanded in a market by the consumers within a given period of time is known as market demand.

Market demand = sum of individual

Types of Demand

1) Price demand:- It is the demand for the product due to the price only / change in the price is known as price demand.

* $D(X)$ is a function of $P(X)$.

$D(X) \rightarrow$ demand

$$D(X) = f[P(X)]$$

$P(X) \rightarrow$ price

2) Income demand:- It is the demand for the product due to the change in the income.

* When the income of a consumer increases, demand for those product which increase more than the income increase known as luxurious product.

* When income increases the demand for those product only increases are called called normal products.

* with in
of those
are known
products.

* Those
uses w
known as

eg $\rightarrow$
box & televis
3) Cross De

It
due to
products.

2 type

1) Substit

2) Comple

sub

gives

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price

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neg

4) Deriv

due t

eg $\rightarrow$

* with increase in income the demand of those products which remain constant are known as necessities/necessary products.

* Those products whose demands decrease with the increase in income are known as inferior product.

eg → some qualities of rice, potato, box television, hand fan

3) Cross Demands:-

It is the demand for the product due to the change in price of related products.

2 types of related product

1) substitute products - eg: tea & coffee

2) complementary products.

substitute products are of same category of products.

In case of ~~sub~~ substitute products there is a positive relationship between price & demand quantity

In case of complementary product the relationship b/w price & quantity ^{demand} is negative.

4) Derived/Indirect Demand:-

It is the demand for the product due to the demand for another product.

eg: Labour demanding copy, machine is indirect.

Factors affecting demand :-

- $D(X)$: demand for product $X = P(X)$ price of X ,
 $P(Y)$ price of Y when Y is a related product of X .
- T (Taste & preference of customer)
- I (Income of the consumer)
- N (No. of consumers residing in the market)
- F (Future expectations regarding the price of the product)

$$P(X) \propto \frac{1}{D(X)}$$

$D(X)$ - quantity demanded.

$$T \propto D(X) \text{ [remain const.]}$$

$$I \propto D(X) \text{ (for normal & luxurious products)}$$

$$N \propto D(X)$$

$$F \propto \frac{1}{D(X)} \text{ (now)}$$

eg of complementary product : ink jet printer & ink cartridge.

Dt :- 1 October 2020

- * If it is expected that price of product is going to increase, the demand will rise.

Law of Demand

other factors remaining constant if the price of the product increases, the quantity demanded for it decreases, & if the price of the product decreases the quantity demanded for it increases.

That means
relationship b/w

Individual

Individual

It is

at various

Price

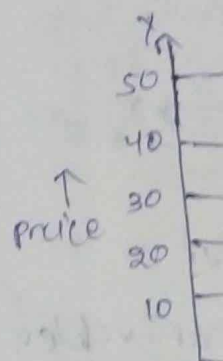
10

20

30

40

50



It

to

b/w

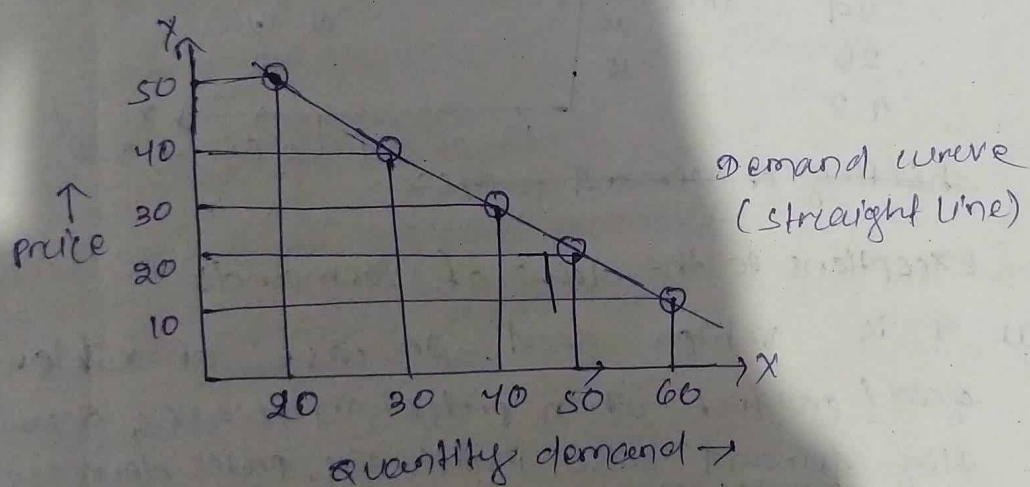
That means there is an inverse relationship b/w price & quantity demanded.

Individual Demand schedule and Derivation of Demand curve

Individual Demand schedule

It is the demand of ~~one~~ costumer at various price in a given period.

<u>Price</u>	<u>Quantity demanded</u>
10	60
20	50
30	40
40	30
50	20



It is downward sloping from left to right showing inverse relationship b/w price & quantity demanded

Market Demand curve:-

Market demand = sum of all the individual demand in the market.

Assume price is const.

Price	Quantity demand A	Q.D. B	Q.D. C
10	50	5	10
20	40	4	8
30	30	3	6
40	20	2	4
50	10	1	2

Market demand

65

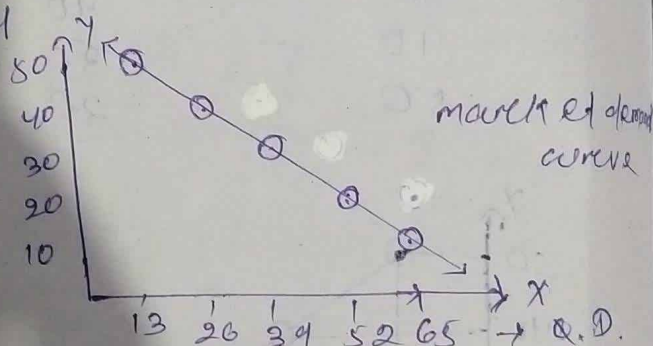
52

39

26

13

price



shifting of demand curve:-

Exceptions to the law of Demand:-

- 1) It is veblen good. In case of veblen good / product when price increases, quantity demand also increases. price decrease Q.D. also decrease.

They are such type of luxurious product & income is directly proportional to demand.

eg. mercedes, designer jewelry [exclusive]

- * They are used for status maintenance.
- * (H) rel? b/w price & demand, also known as snob or demonstration effect.

DE-5042020

d(a)

-b →

P(S)

d(a) =

A=10

b=2

P=2

d(a) =

* If price

Elastic

It is

the

to change

change

→ 3 factors

of demand

* change

* Income

* price

According

1) Price

2) Income

3) Price

Price

in

to the

for

proportion

proportion

DE: 5 Oct 2020

$$d(a) = A - bP(s)$$

$-b \rightarrow$ negative slope $A \rightarrow$ const. term

$P(s)$ = price of product

$d(a)$ = demand of the product

$$A = 10$$

$$10 - 2 = 8 \rightarrow d(a)$$

$$b = 2$$

$P(s)$ initial is 1

$$P = 2$$

$$d(a) = 6$$

* If price increase, demand decreases

Elasticity of demand:-

It is the degree of responsiveness ~~due~~ to in quantity demand of a product due to change in price, change in income & the change in the price of related product.

$\rightarrow$ 3 factors affecting degree in change of demand $\rightarrow$

* change in price,

* income,

* price of related product

Accordingly there are 3 types of elasticity

- 1) Price elasticity of demand
- 2) Income elasticity of demand.
- 3) cross elasticity of demand

Price Elasticity

It is the degree of responsiveness in quantity demand of a product due to the change in price.

Formula for calculating it is:-

$$\frac{\text{Proportional change in } \cancel{\text{Q.D.}} \text{ quantity demand}}{\text{Proportional change in price}}$$

$$E_d = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q}$$

= $\frac{\text{proportionate change in quantity demanded}}{\text{proportionate change in price}}$

$$= \frac{\frac{\Delta Q}{Q} \times 100}{\frac{\Delta P}{P} \times 100} = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q}$$

Q) As a result of 10% fall in the price of a product, its demand raises from 10 to 12 unit. find out price elasticity of demand.

$$\frac{\Delta P}{P} \times 100 = 10$$

$$\frac{\Delta Q}{Q} \times 100 = \frac{2}{10} \times 100 = 20$$

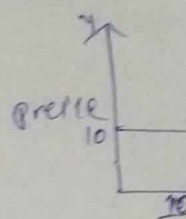
$$\frac{20}{10} = E_d = 2$$

* there is no negative price elasticity of demand.

There are 5 types of price elasticity of demand.

- 1) perfectly elastic
- 2) perfectly inelastic
- 3) relatively elastic
- 4) relatively inelastic
- 5) unitary elasticity

① perfectly elastic price elasticity of demand
if there is a negligible change in price leads to a greater change in quantity demanded.



$$E_d = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q}$$

$$E_d = \infty$$

② Perfectly inelastic which

② Perfectly elastic which

price demand said to

$$E_d =$$

$$E_d = 0$$

eg → so

③ Relatively elastic which

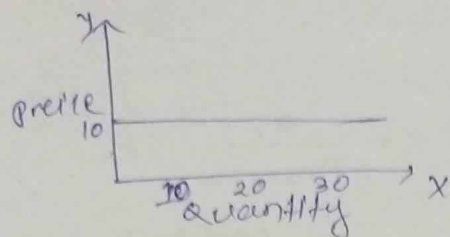
in price

onate

price

elastic

eg →



$$E_d = \frac{\text{proportionate change in Q.d.}}{\text{proportionate change in price}}$$

$$= \frac{\Delta Q}{Q} \times 100}{\frac{\Delta P}{P} \times 100} = \infty \quad \text{K (some value)}$$

$$E_d = \infty$$

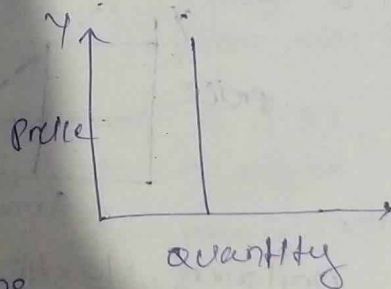
① there is no such product found which comes under perfectly elastic one.

② perfectly inelastic

when a greater % change in price leads to no change in quantity demanded, then the price elasticity is said to be perfectly inelastic

$$E_d = \frac{\frac{\Delta Q}{Q} \times 100}{\frac{\Delta P}{P} \times 100} = 0$$

$$E_d = 0$$

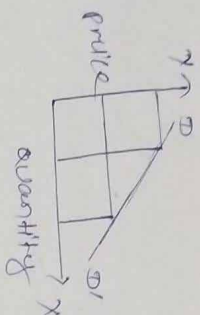


eg → salt, rice, medicine

③ Relatively elastic $[|E_d| > 1]$

when a given proportionate change in price leads to a greater proportionate change in quantity demanded, the price elasticity is said to be relatively elastic.

eg → all luxurious product



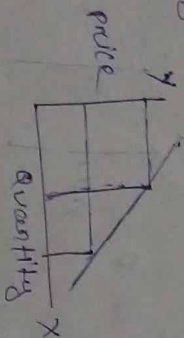
Relatively inelastic

A given proportionate change in price leads to lesser / less than proportionate change in ~~price~~ quantity demand.
20% change in price leads to 10% change in quantity demand.

$$Ed = \frac{20}{10} = 0.5$$

$$Ed < 1$$

eg vegetable, food, petrol



unitary Elastic

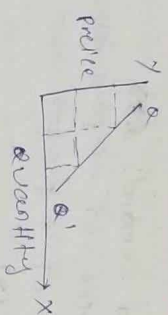
A given proportionate change in price leads to equal proportionate change in quantity demand.

$$Ed = \frac{10}{10} = 1$$

$$Ed = 1$$

10% increase in price leads to 10% decrease in quantity demand.

eg all normal products



Dt :- 804 2020

As a result of 10% fall in price of product, its demand rises from

eg At a price Rs 40/unit, the quantity demand of the product is 500 units. when its price falls by 10%, its quantity rises to 550 units. calculate its price elasticity of demand.

$$\frac{\frac{\Delta Q}{Q}}{\frac{\Delta P}{P}}$$

$$= \frac{50}{500} \times \frac{1}{10} \times 100$$

= 1 (unitary price elasticity)

eg price elasticity of demand is (-3). units of this good are brought at price of Rs 8/unit. How many units will be bought at Rs 9.

$$\frac{\frac{\Delta Q}{Q}}{\frac{\Delta P}{P}} = -3$$

$$\frac{\frac{\Delta Q}{40}}{\frac{1}{8}} = \frac{\Delta Q \times 8}{40} \times -3$$

$$\Delta Q = 9, + \Delta Q = 40 - 15 = 25$$

Income elasticity of demand: ~~It~~

It is the degree of responsiveness in quantity demanded of a product due to the change in income.

Formula for calculating income elasticity:

proportionate change in quantity demanded
proportionate change in income

$$= \frac{\frac{\Delta Q}{Q} \times 100}{\frac{\Delta Y}{Y} \times 100}$$

$$I.E. = \frac{\Delta Q}{\Delta Y} \cdot \frac{Y}{Q}$$

* There is negative income elasticity of demand, it implies that product is an inferior product.

If the income elasticity is (+) or greater than 1, the product is a luxurious product.

If income elasticity is 1, the product is a normal product.

If income elasticity is 0, the product is necessity.

Q	tea	10 → 50	
P	coffee	50 → 60	$\frac{10}{50} \times \frac{10}{60} = \frac{1}{30}$

Cross elasticity of demand:

It is the degree of responsiveness in quantity demanded of a product due to the change in the price of the related product.

Formula

case
proportionate change in quantity demanded of a product
the related product

cross elasticity = proportionate change in price of the related product

* If cross elasticity is (+), the product is a substitute product.

* If cross elasticity is (-), the product is a complementary product.

substitute: Both price and demand increase of both.

complementary: If one product's demand increases, quantity demanded of another decreases.

~~It is good goods~~

substitute
complementary

$$\frac{\Delta Q_a}{\Delta P_b} \times \frac{P_b}{Q_a} = \text{price elasticity}$$

$$\frac{\Delta Q_a}{\Delta P_a} \rightarrow \text{price elasticity}$$

Factors affecting price elasticity of demand:-

- 1) Nature of the commodity (X)
- 2) Availability of substitute (X) demand elasticity
- 3) proportion of income spent & demand elasticity
- 4) Habits and fashions
- 5) The state of joint demand -
for eg. we demand both products
elasticity will depend upon other product
- 6) several uses of product
no. of uses & demand elasticity of product
- 7) period, short period more elasticity
- 8) level of fluctuation $\rightarrow$ more fluctuation more elasticity

Supply of the product

supply means availability of goods in market for sale.

But in economics it has a diff. meaning. It refers to the quantity of a product that the seller offers for sale at a given point of time & in a specific market.

70 Perfect supply is (things)

- 71 Producers must ~~produce~~ ^{have} some goods & services.
- 72 Producers must have wish to sell the

product at market ^{want to} sell it at prevailing price.

Factors affecting supply:-

The quantity supplied of a commodity has a positive relation with the price, but there are some other factors which determine the extent of supply:-

- supply of product X, $SC(X)$ is a function of $PC(X)$ & price of product X ~~is a function~~ $\rightarrow$
- 1) price of related product, $PC(Y)$
- 2) price of related product, $PC(Y)$
- 3) $N \rightarrow$ no. of farmers producing product.
- 4) $I \rightarrow$ Input price
- 5) $G \rightarrow$ govt. policy, ^{after} ~~tax~~ & subsidies
- 6) $F \rightarrow$ future expectation regarding the price of the product.

Technology (T)

- 1) $PC(X)$ & $SC(X)$
- 2) $PC(Y)$ & $SC(X)$
- 3) N & $SC(X)$
- 4) I & $SC(X)$
- 5) T & $SC(X)$

Primitive technology $\rightarrow$ production less $\rightarrow$ supply less

If the production is subsidised the production is more
if taxation ~~is~~ imposed on production is more, supply will be less.

Substitution & SCX) & Tax & $\frac{1}{SCX}$

F & $\frac{1}{SCX}$

If it is going to be high SCX will be ~~more~~ less now.

Law of supply :-

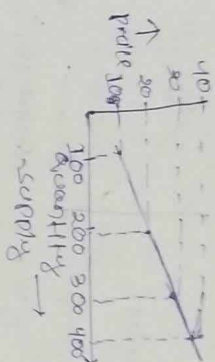
other factors remaining constant, if the price of the product increases, then the quantity supply also increases. If the price of the product decreases, the quantity supply will also decrease.

Individual supply schedule

It is a tabular statement which provide statement of supply of diff. commodities at diff. prices.

It is a tabular representation of price & quantity supply of one individual at a given period of time.

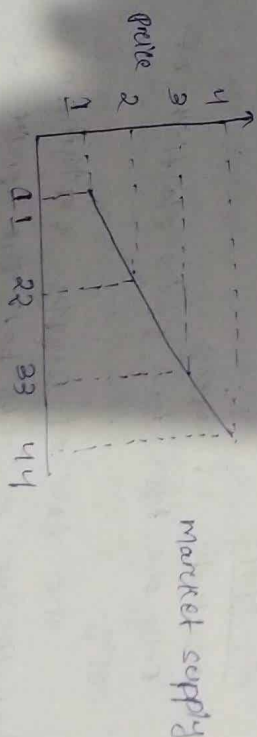
price	quantity supply
10	100
20	200
30	300
40	400



It shows a positive slope. It shows at higher price the supply will be high.

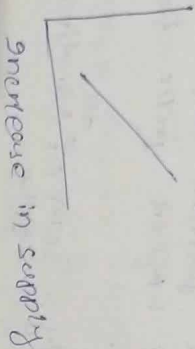
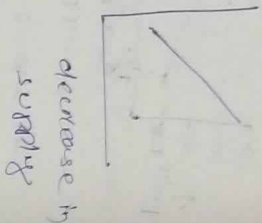
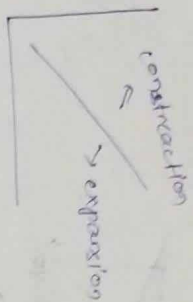
Market supply :- sum of all individual supply

price	Q.S (A)	Q.S (B)	Market supply
1	10	1	11
2	20	2	22
3	30	3	33
4	40	4	44



shifting of supply curve

expansion & contraction of supply curve due to change in other factors which are assumed to be const.



- * if technology improves $\rightarrow$ curve moves to right.
- * if input price decreases $\rightarrow$ curve moves to right.
- * price increase $\rightarrow$ curve expansion (outward shift).

dt: 12/10/20

Elasticity of supply:-

only one type i.e., price elasticity of supply.

The law of supply explains a

qualitative relationship between the price & supply, whereas the elasticity of supply refers to a quantitative relationship.

between the price & quantity supplied. Elasticity of supply is the degree of responsiveness of supply of a

commodity to a change in its price.

5 types

- 1) Perfectly elastic:- (oo) curve horizontal. negligible change in price leads to great change in supply.
 - 2) Perfectly inelastic:- (0), curve vertical. greater % change in price leads to no change in supply.
 - 3) Relatively elastic:- given proportionate change in price leads to a greater proportionate change in supply (>1)
 - 4) Relatively inelastic:- given proportionate change in price leads to lesser proportionate change in supply (<1)
 - 5) Unitary elastic:- A given proportionate change in price leads to an equal proportionate change in quantity supplied ($=1$)
- * No negative price elasticity of supply.
- Proportionate change in quantity supply elasticity = $\frac{\text{proportionate change in price of supply}}{\text{proportionate change in quantity supply}}$

$$s = \frac{\Delta Q_s}{Q_s} \times 100$$

$$\frac{\Delta P_s}{P_s} \times 100$$

$$s = \frac{\Delta Q_s}{\Delta P_s} \times \frac{P_s}{Q_s}$$

egs 9 farm supplies 200 units of a product at a price of Rs/units, when price changes it supplies 100 units less. price elasticity of supply is 2.5. calculate price after change.

$$\frac{-100}{200} \times \frac{P}{\Delta P} = 2.5$$

$$= 2.5 \times \frac{1}{\Delta P} = 2.5$$

$$\Delta P = 1$$

$$P_2 = 4 \text{ Rs.}$$

egs 9 A farmer supplies 100 units of a product at a given price. when price falls by 10% then the farmer supplies 70 units. Price elasticity of supply is 3. Calculate the given price

$$\frac{-30}{100} \times \frac{P_s}{-1} = 3$$

$$= \frac{3}{10} \times \frac{P_s}{-1} = 3$$

$$\frac{P_s}{10} = 10 - \text{original price}$$

$$P_s = 10$$

Price determination in a perfectly competitive market:-

perfect competition
large no of buyers & large no. of
sellers & a single prevailing price,

imperfectly competitive is
varying prevailing price.

* No perfectly competitive market in
real world, though price market is
concentrated in this category.

13/04/20 an a perfectly competitive market

the price is decided where the
quantity demanded = quantity supply, which
is called equilibrium price.
or market demand = market supply

Price	Q.d	Q.s	
10	100	60	40 excess demand
20	90	70	
30	80	80	
40	70	90	
50	60	100	
here eqm price = 30			

$$\text{eqm Q.d} = \text{Q.s} = 80$$

situation 1 & eqm price $\rightarrow$ excess demand, when

Q. demand is more & Q.s is less

② price $\uparrow$ eqm price $\rightarrow$ excess supply, when
price = 50, Q.d = 60 but Q.s is 100, so, excess
supply is 40.

* O/c to theory, price $\frac{1}{\text{demand}}$ & supply

At price 20 units of buyers are of
90, 20, so they will try to get the
product by paying high amount hence

price increases.

when price increase the demand decreases as ^{some} people won't want to buy that product at higher price.

As to the supply when price increases, supply also increases.

In excess supply condⁿ, in order to sell the excess amount the seller will reduce price, as a result demand will increase. Thus it will again reach the eq^m price.

③ eq^m price when quantity demand = quantity supply.

* supply & demand are self adjusting & they tend to adjust in such a way that it will reach eq^m price.

$$D_a = a - bP_a$$

(-b) slope of demand curve

D_a = demand of product a

P_a = price of product a

$$S_x = a + bP_x$$

S_x = supply of product a

Q. suppose $Q_A = 40 - 2P$

$$Q_D = 30 + 3P$$

find out the market demand

$$Q_A + Q_D = 70 - 3P$$

market demand = sum of individual demand

Q. suppose,

$$Q_S = 10 + P$$

$$Q_D = 20 + 2P$$

$$\text{market supply} = 30 + 2P \quad (Q_S + Q_D)$$

Q. price, (keep market demand = market supply)

$$\text{Q. suppose } Q_A = 70 - 2P$$

$$Q_B = 200 - 4P$$

$$Q_C = 20 - 0.5P$$

market supply = $40 + 3.5P$, find out price, market demand &

$$\text{market demand} = Q_A + Q_B + Q_C$$

$$= 290 - 6.5P = 290 - 6.5 \times 25$$

$$\text{So, } 290 - 6.5P = 40 + 3.5P$$

$$250 = 10.0P$$

$$P = 25 \text{ Rs}$$

market supply = 40 + 3.5(25)
= 121.5

at: 3rd Nov 2020

Production

An economic by production means the creation of utilities (the ~~total~~ ^{want} satisfying capacity of a product $\rightarrow$ utility).

production means the transformation of input into outputs.

Inputs are also known as the factors of production.

4 factors of production are there:-

- 1) land
- 2) labour
- 3) capital
- 4) organisation/entrepreneur

suppose output is Q , ~~the~~ then $Q = f(L, land, L, ca, org)$

production function:-

it can be discussed in period of time, 1) short run
2) long run

1) short run is a period of time in which we can keep some factors as fixed and vary the other factors. those factors whose amount are kept fixed in short run are known as fixed factors. eg 1) land & 2) organisation.

those factor whose amount are variable in the short run are known as variable factor.

eg 1) labour & capital.

$Q = f(L, labour)$

suppose $100 = f(1, 10)$

$\rightarrow \frac{100}{1} = 100$ (impact on output)

short run production function is also known as the law of variable proportion

2) long run:- long run is a period of time in which all the factors of production are variable.

suppose $Q = f(L, 1, 12)$

12 $\rightarrow$ land

land $\rightarrow$? $\rightarrow f(2, 14)$

long run is also known as the law of returns to scale.

The law of variable proportion:-

suppose $Q = 100 = f(10, 10)$

10 $\rightarrow$ land
10 $\rightarrow$ labour

proportion 1:1

land & labour = 1:1

$f(10, 20)$

proportion = 1:2, (the impact on output with the varying proportion).

it is called so, ~~that's~~ because proportion b/w fixed factor & variable factor will change.

Why short run is called as the law of variable proportion?

Because in short run we keep some factors fixed while others vary while doing so, the proportion between fixed factors & varying factor change. As we discuss the impact of this on the output in the short run so we call short run as law of variable proportion.

60% of population in agriculture are engaged in agriculture, but we have not that much land. Since the land is fixed & labour increase, the proportion varies, the output initially will increase till a given period of time i.e., 1^{st} max. / proper utilisation after that it will decrease due to more wage contribution. This is called disguised unemployment.

Labour
 (1, 1) - 60
 (1, 2) - 70
 (1, 3) - 80
 (1, 4) - 80
 (1, 5) - 70
 the extra labour will take wage (1 unit) from output.

5/7/2020
 Assumptions of the law of variable proportion:-

1) The technology remains constant i.e., no technological change
 2) There must be only one factor which is variable in nature & others factors are constant.

3) It is a short run production function

Total product (not negative)

suppose 3 fixed factor

land labour
 (1, 2, 3) - 100 unit

T.P. = 100 unit

Average product = $\frac{\text{total product}}{\text{no. of factors of production}}$
 (can be negative)

$$= \frac{100}{3} = 33.33 \text{ units}$$

marginal product's change in the total product (marginal)
 (can be negative) change in no. of units of variable factor

for eg →
 initial land labour product
 1, 2, 3 - 100 unit

marginal product of last labour =

$$\frac{100 - 100}{3 - 2} = 100 \text{ units.}$$

marginal product:-

product produced by additional / last units of factors of productions.

→ Total product is the addition of marginal product.

capital	M.P.	T.P.
1	10	10
2	8	18
3	6	24
4	4	28

These laws are discussed in 3 stages →
 1st stage of increasing return
 2nd stage of diminishing return
 3rd stage of negative return

Increasing Return:-

Land	Labour	T.P.	M.P.
1	1	2	2
1	2	5	3
1	3	9	4
1	4	12	3
1	5	14	2
1	6	15	1
1	7	15	0
1	8	14	-1
1	9	13	-2

Law of diminishing marginal utility →
 marginal utility → amount of satisfaction
 no. of person constant.

Capital :-
 money that is invested in the organisation

Increasing Return (1st stage)

A.P. increases, T.P. increases in an increasing rate.
 A.P. increases, M.P. increases, & marginal product ends where average product = marginal product
 * M.P. max in 1st stage & end stage of diminishing return

M.P. decreases & becomes zero

A.P. decreases

T.P. increases at an decreasing rate

suppose M.P. of labour

	1 st Labour	2 nd Labour
2	2	2
3	5	5
4	9	9
3	12	12
2	14	14

increases at increasing rate

Point of inflection :-

The point upto which total product increases at increasing rate

when marginal product is 0, total product is max → here the 2nd stage ends

2nd stage →

T.P. decreases as marginal product became negative

A.P. decreases, remains positive

so, this is known as stage of negative return, because marginal product of variable factor is negative.

A rational producer will always choose 2nd stage because in 1st stage the fixed factor is not properly used. eg: land here.

→ In 2nd stage the variable factor is over used. eg: labour here.

Q1: 10 Nov 2020

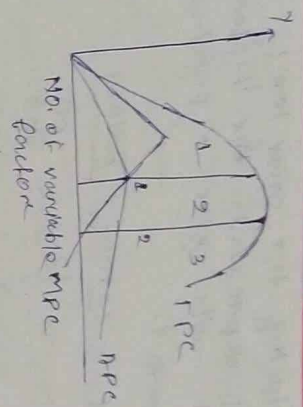
Labour	M.P.	T.P
1	10	10
2	10	20
3	10	30
4	10	40
5	10	50

1st & 2nd stage both the functions are completely used & T.P. is max.

50 = 10 - Average product

Curve
→ point of inflection → upto this point T.P. increases at an increasing rate.

1 → 10 where T.P. reaches the point of inflection, M.P. is max
10 → 20
20 → 30
30 → 40
40 → 50
then M.P. starts decreases and becomes 0, where T.P. becomes max.



1st stage

T.P. & M.P. both max.

Q1: 16 Nov 2020

Long run production function

* The proportion b/w all the variable factors remains const.

eg: Land	labour
10	10
20	20

Assumption

* All the factors will vary but the proportion b/w all the variable remains const.

* If we all are dealing factors → uncertainty

objective & its implication output is discussed.

* Long run production function is known as to study because we will discuss the returns on output (while the input change in a given proportion) or its impact on returns on output.

Assumption:

also known as equal product curves.

* Equal product curves represent all those factor combinations which give

capable of producing the same level of output or it represents the factor combination which will give a producer equal same level of output.

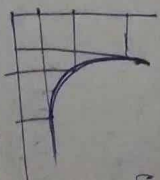
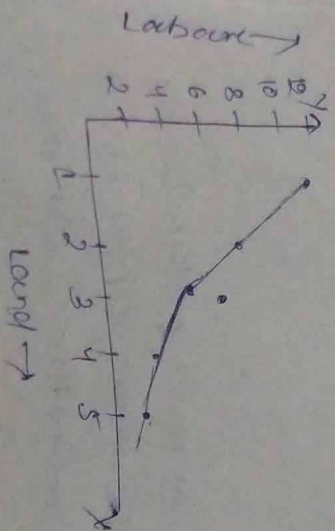
Land	Labour	output
1	12	100
2	8	100
3	5	100
4	3	100
5	2	100

the rate of marginal rate of factor substitution, diminishing marginal rate of substitution, the rate at which one product is substituted by other.

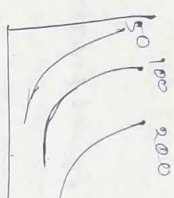
* budget of producer is const, so after buying one, he has to forgo some units of other factor while buying another after buying land 1, 12 labour 1 if the buyer wants to buy another 1 land then the amount of labour is 8, he/she must forgo 4 units of labour.

* the rate of sacrificed unit goes on decreasing.

* producer is a rational producer.



actual shape
convex
equal product curve



Right → more
Left → less

isoquant map (more than one curve)

3 laws of Return to scale:-

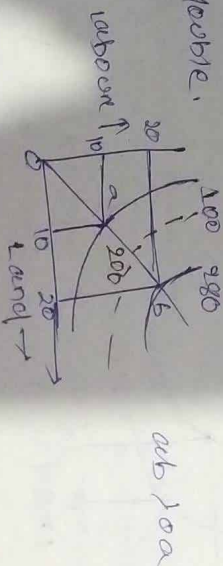
- ① ~~constant~~ increasing Returns to scale.
- ② constant Return to scale
- ③ diminishing Return to scale

Increasing Return to scale

when a given proportionate change in inputs leads to a greater proportionate change in output, it is known as increasing returns to scale.

Land	Labour	output
10	10	100
20	20	280

The input is 2 times, output is more than double.



Factors responsible

* Some factors are available in fixed amount, we can not change its size, when we go on increasing other factor the production will increase more than change in variable increasing labour (joint variability of factor)

DATE: 19 NOV 2020

National Income

* It comes under macroeconomy.

* National Income is the money value of all the goods & services produced in a country within a given period of time.

defn It is the labour & capital of the country acting upon natural resources produces commodity including all the ~~the~~ ^{goods} ~~at goods~~ ^{commodities} including services of all kinds. * addition of all the income of people residing in an economy.

* Income from abroad.

GMR → gross National product

money value of goods & services produced within a economy including income from abroad.

2-type GNP at market price (GNMP)
GNP at factor cost (GNPFC)

GNP = GMR

$$* \left[\frac{C+I+G}{C+I+G} + (X-M) \right]$$

It is also called money value of goods & services produced within a economy during a given period of time which is also calculated by $C \rightarrow$ consumption expenditure

by $I \rightarrow$ investment expenditure

$G \rightarrow$ govt expenditure

$X \rightarrow$ export $(X-M) \rightarrow$ net factor Income

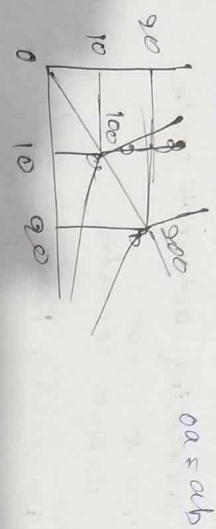
$M \rightarrow$ Import from abroad

National Income is calculated:-

- ① product method } mostly used in India
- ② Income method }
- ③ Expenditure method }

* Const. Return to scale
the output will change in equal proportion as that of input

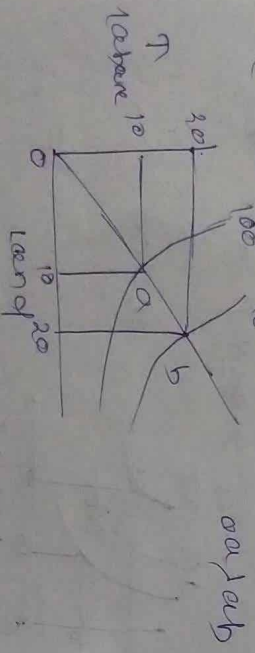
$$100 = (10, 10) \\ 200 = (20, 20)$$



diminishing Return to scale

The output will change smaller proportion as compared to proportionate change in input.

$$\text{Land/Labour} \\ (10, 10) - 100 \\ (20, 20) - 180$$



GNPFC

$$\boxed{\text{GNRMP} - \text{Indirect tax} + \text{Subsidy} = \text{GNRFC}}$$

Factor cost $\rightarrow$ Land, Labour

$\rightarrow$ Indirect (wholesaler, retailer pay indirect tax)

tax $\rightarrow$ direct

$\rightarrow$ indirect (cost) $\rightarrow$ goods & services pay

subsidy $\rightarrow$ given by govt. to people.

GNPFC $\rightarrow$ Net national product

$$\boxed{\text{GNPFC} - \text{depreciation} = \text{NNRP}}$$

2 types: NNRMP & NNRFC

$$\boxed{\text{NNRMP} = \text{GNRMP} - \text{depreciation}}$$

* Return & return changes $\rightarrow$ depreciation

$$\boxed{\text{NNRFC} = \text{NNRMP} - \text{Indirect tax} + \text{subsidy}}$$

GNP

GNP $\rightarrow$ Gross Domestic Product

$$\boxed{\text{GNRMP} - \text{net factor income from abroad}}$$

$$\boxed{C + I + G = \text{GNP}}$$

It is the value of the country's output upon natural resources produced annually or certain net aggregate of commodities.

GNP now $\rightarrow$ ~~23.9%~~ 23.9% (gross)

apport from agriculture all sectors have

negative GNP

(boom) or

economy is never lost, it has inflation as well as recession, in boom, people don't have the income / purchasing power to produce goods & services,

along the well & fill it up $\rightarrow$ 1940, Britain, Korea.

24.03 Nov 2020

GNPMP

It is the money value of goods & services produced within the country. ~~within a~~ during a given period of time.

* Difficult to GDP & GNP:

GNP we do not include the net factor income from abroad.

$$\boxed{\text{GNRFC} - \text{Factor cost}}$$

$$\boxed{\text{GNRMP} - \text{Indirect tax} + \text{subsidy} = \text{GNRFC}}$$

* because govt. expenditure is there, we deduct indirect tax.

NNRMP $\rightarrow$ Net Domestic product at market price.

$$\boxed{\text{NNRMP} = \text{GNRMP} - \text{depreciation}}$$

2 aggregates of National income

NNRFC $\rightarrow$ National income in country

~~next to the~~ methods to measure national income:

* GNP $\rightarrow$ gross value added at market price

3 methods

① product or value added method

② income method

③ Expenditure method.

- * the economy is divided into 3 sectors, to
 ① primarily calculate national income, to
 ② secondary value added method,
 ③ tertiary

At the goods & services

National income :-

value of goods & services, primarily + (goods & services)

+ (goods & services) tertiary

- [Intermediate consumption (primary + secondary + tertiary)]
- ① fishery, mining, quarrying, agriculture etc.
 - ② construction, infrastructure, electricity, manufacturing etc.
 - ③ service sector, hotel, tourism, banking.

intermediate consumption, which are again
 goods & services which are again
 used in production system/process.

- * It is also known as value added method as value of goods & services are added.
- * In order to prevent the double counting intermediate consumption is deducted.

problem / limitation

- ① Because most of the sectors are not manufactured yet, market system, changes of double accounting, whether the product is intermediate or final it is difficult to say.
- ② Market are not systemised yet. Apart from it this method is widely used to measure national income.

* our economy is unsystematic, & we do not want to accept changes or reforms, thus having problems in national income. MSP + minimum support price, which is fixed in 1970s, the farmers are selling at a price much lower.

National income in income method
 Net Rent + wage + interest + profit + income of factors of production + income which is from mixed goods, i.e., the extra money ~~from~~ earned by the factors of production / people engaged in dual economy. $\Rightarrow$ is = national income

expenditure method

$$C + I + G + (X - M)$$

26 Nov. 2020

* Income method

~~income of factors of production~~ (income of people residing in an economy.)

$\Rightarrow$ mixed income of self employed + Rent + wage + interest + profit = national income

* expenditure

expenditure of a people + expenditure of govt = national income

CSD $\rightarrow$ central statistical office ~~calculate~~ measure the national income of our country

Classification
economy

Barter system
(agents & services are exchanged by another agent & services)

Limitation of income method
people do not show their exact income
i.e. not paying proper tax.

* In a manufacture & construction, expenditure method is used to calculate national income.

Income
Calculation of expenditure method is easy.
It is difficult to maintain a record of it in case of private organisation.

* Income & products method are equally used.

Cost & Revenue

Cost
The expenses / expenditure incurred for producing a product is cost of a product.

Cost function

Functional relationship b/w the cost & output is known as the cost function.

- * cost is the function of the output.
- * types of cost funcⁿ:
- ① short run cost funcⁿ
- ② long run cost funcⁿ

Short run
2 types of cost
① fixed cost

② variable cost
As the factors of production are divided into two types so, total cost = total fixed cost + total variable cost.

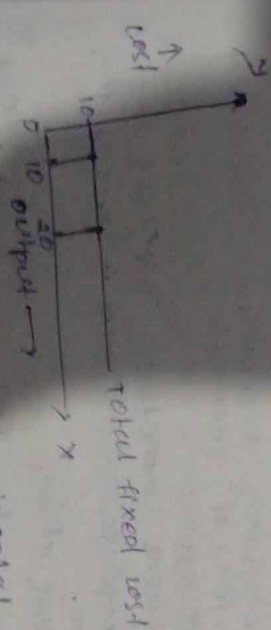
* In short run total fixed cost is incurred in the factors of production that are fixed in nature.

* remain fixed throughout the production time in a given period of time.

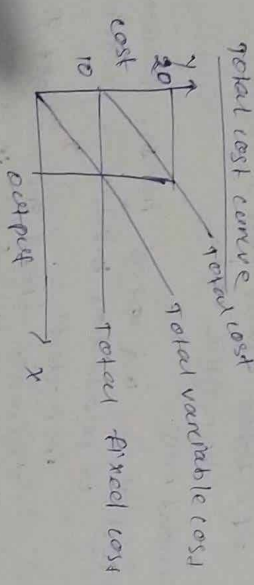
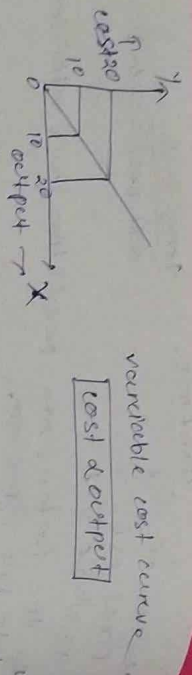
total variable cost:

* varying with the output.
* cost incurred in the factors of production that are variable in nature.

from variable cost
① raw material cost



* Total fixed cost is horizontal in nature.



total cost starts from 0 and is :-
because it is the collection of
fixed cost & variable cost

* Total variable cost is also known as prime cost.

prime cost are those cost directly proportional to output

* The fixed cost is also known as supplement cost.

① monetary expenses:-

* The monetary expenditure incurred in various functions of production.

② real expenses:-

apart from monetary expenses the real expenses incurred in production are physical effort eg physical effort

③ opportunity cost:- cost of next best product / alternative

which is forgone to produce a product is known as opportunity cost

suppose 2 products
X → production cost is 100 & producer get 500
Y → production cost 100 & producer get 300

he will choose X.

(X) opportunity cost = 100 (Y) which is forgone to produce X as X is more profitable than Y.

eg. an assam tea is produced most. But due to assam's atmosphere rice is the most suitable product. but rice is the most profitable, the cost of rice is forgone to produce tea. is the opportunity cost for tea.

explicit cost → outside cost
those cost which are borne by the producer which are outside the organization

implicit cost → inside cost
borne by producer / owner

* If the tractor is hired by producer it is explicit, but if the producer owns the tractor, then it is implicit cost.

December 1990 Suppose X = 50 (Total production cost)

① Total cost → cost of production of 5 unit = 100

cost incurred in the factors of production.

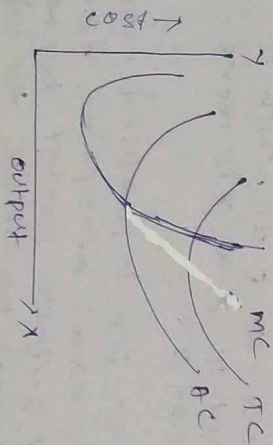
② Average cost = $\frac{\text{Total cost}}{\text{no. of unit}}$ avg cost = $\frac{100}{5} = 20$

③ Managerial cost :-
change in total cost
change in no. of unit

marginal cost = $\frac{\Delta TC}{\Delta Q}$ (per unit cost of production)

if unit is produced in 1000s
cost is produced in 1000s

marginal cost = $\frac{20}{1} = 20$ Rs

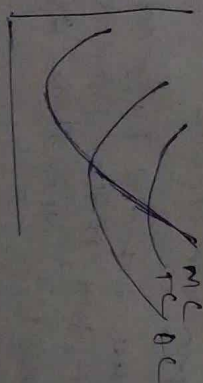


The marginal cost curve intersects both AC & MC curve

* At the intersection avg. cost is min, marginal cost is already min.

* curves are U-shaped as cost ~~decreases~~ decreases, then reaches min & then increase with increase in output.

* MC, MC, TC is just opposite of MP, MP, TC.



* 1st marginal cost decrease & increase then.

* at point of intersection AC is min

* The shape of TC depend upon MC, as MC increases TC increases & vice versa.

* At intersection, min AC & min TC respectively

Revenue

* revenue is the amount of money received after selling the product.

$$\text{Profit} = TR - TC$$

TR = total revenue
TC = total cost

* increasing part, where $TR > TC$ cost expenditure incurred.

Total Revenue = addition of Revenue earned from all the unit.

unit	1	2	3	4	5	6	7
TR (Rs/Unit)	12	24	36	48	60	72	84
TC (Rs/Unit)	10	20	30	40	50	60	70
TR	12	24	36	48	60	72	84
TC	10	20	30	40	50	60	70
TR - TC	2	4	6	8	10	12	14

* tendency of seller to sell the product at higher price initially

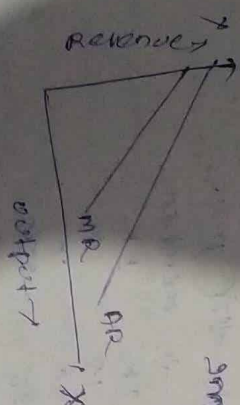
* marginal revenue & total revenue change in no. of unit

* average Revenue = total revenue / total no. of unit

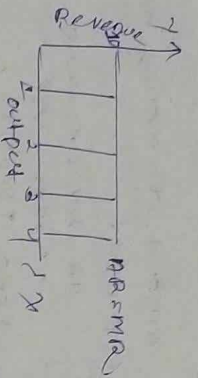
* total Revenue

* AR will be higher than MR

the price of the product is generally known as not perfectly competitive market.



perfectly competitive market $\rightarrow$ $AR = MR$



perfectly competitive
 $AR = MR$

TR	MR	AR	Unit	Q
10	10	10	1	
20	10	10	2	
30	10	10	3	
40	10	10	4	
50	10	10	5	
$AR = MR$				

because each and every unit will be sold in the same price

Q: 1-1/12/2020

Marginal Revenue →

Revenue per head / received after selling per no. of units.

8 unit → 100 Rs $AR = \frac{100}{8} = 20$

6 unit → 120 Rs

$MR = \frac{120-100}{6-5} = 200$

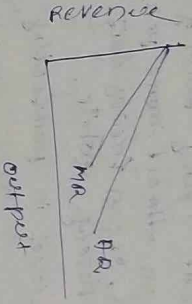
average of money received → average revenue
addition of marginal revenues → total revenue
perfectly competitive market
a single price prevail in market
perfectly competitive market

each unit sold at diff price.

* always present in members.

no. of unit	MR	TR	AR
1	12	12	12
2	8	20	10
3	5	25	8.33
4	3	28	7
5	2	30	6

AR, MR (perfectly competitive market)
price of the product



both curve are downward sloping showing the negative relationship b/w Revenue & output.

Market structure

Market where the potential buyers & sellers will be brought together & they are making the competitiveness buying or selling

ability to produce goods & services → seller
ability to purchase goods & services → buyer
* on the basis of competition / no. of seller
2 types of market

① perfect competition

price is decided by demand & supply of the product

* market structure where no individual among the producers within the market.

* large no. of buyers & sellers.

* the farming produce is a price taker not a price maker

* everybody is aware about the perfect knowledge about market situation.

* price is already decided, the farmers

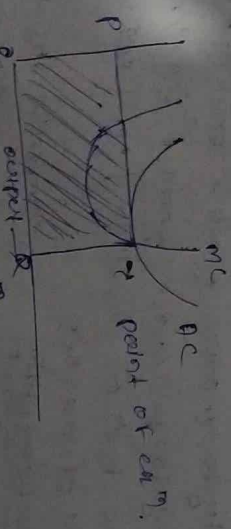
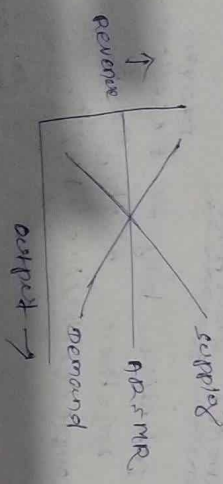
will not decide it.

* homogeneous product is product of same quality, shape, size is available in many

* free entry & free exit $\rightarrow$ any farmer getting increasing profit, it can attract more farmers & if facing any losses, it can freely exit.

* each resource is freely mobile.

how farm output is decided in perfecting competitive market?



secondly for farmer eqm

MR=MC

② MC must be rising at the point of eqm

* in perfect competition resources are completely used, because the product is produced & sold at min^m point of average revenue $\rightarrow$ proper utilisation of resources.

* normal profit $\rightarrow$ all gets.

* As a farmer is only a tiny unit, it in case of large no. of buyers & sellers so it can't decide the price.

* optimum utilisation of resources in perfect competitive market.

At: 4 Dec 2020

Imperfect competition

Monopoly $\rightarrow$ mono-single, poly-seller

It is such type of market where a single seller & large no. of buyers are there.

① single seller & large no. of buyers

② strong barriers to entry into the market

③ no free entry & exit

④ the product does not have close substitutes

⑤ cross elasticity b/w the monopoly sellers is zero.

⑥ product & product of others sellers is zero.

eg: Indian railways

⑦ types of monopoly (3 types)

① pure simple monopoly

② pure monopoly

③ disruptive monopoly

④ simple monopoly

⑤ which changes min^m profit which does not attract new farmers into the market.

⑥ not attract new farmers into the market.

⑦ that type of monopoly is simple monopoly.

⑧ pure monopoly is when only single seller at the product is there in market.

eg: patent of company

patent right

disruptive monopoly

that type of monopoly seller, which sell same product to diff buyers at diff prices.

on who changes diff price to diff prod
for same product.
How to ~~measure~~ monopoly power?
comp? Levens comp?

MR-MR = monopoly power

MR → average revenue

MR → monopoly revenue

cross elasticity is 0, pure monopoly (strong monopoly)

is 1, no monopoly.

→ generally in perfect comp. each farm only get normal profit, few farm exit, if they face loss.

→ monopoly seller can earn super normal profit as there is strong barrier to entry in the market & a single seller is there.

② Monopolistic competitive market

today's market is coming closer monopolistic type of comp. market because large no. of buyers & sellers are there.

comp. characteristics

product differentiation is ~~the~~ product is diff from one to other, so the price is different from one to other.

diff in terms of colour, shape, size & fragrance → product diff.

→ packaging system

→ advertisement play a very imp role in demand of the product in monopolistic type of competitive market

et. throat competition is there is heavy competition in the market, if u could not

able to sustain, u will be wiped out.
AS, MNCs are there in market.
* max 80-90% small scale industries are sick industries.

features

→ large no. of buyers & seller

→ product differentiation, so the price diff.

→ advertisement plays imp. role

→ demand of diff product is diff.

→ cut throat comp.

③ Oligopoly

oligo → few

poly → sellers

→ few sellers & large no. of buyers are there. (few farms)

in order to prevent more comp.

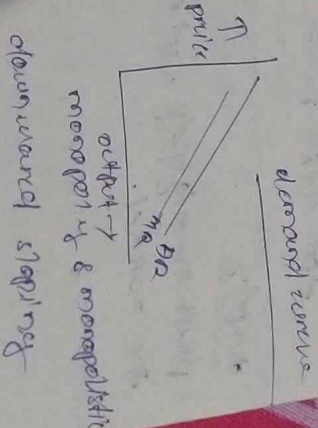
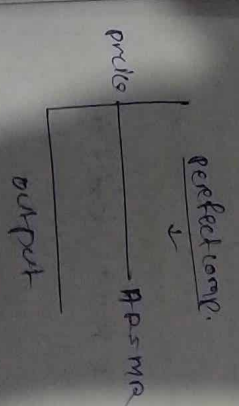
they use form merger / collaboration.

* market structure having strong barrier to entry.

* price & output are not fixed, oligopoly has

face indeterminable demand curve

* demand curve ~~has~~



* no fixed relationship b/w output & price in case of oligopoly.

demand curve $\rightarrow$ revenue curve.
 * the price & output of one farm depend upon the price & output of other farms, so demand curve is interdependent.

* mutual interdependence: one farm has to depend on other farms to face the market.
 two types of oligopoly farms

1) cartel $\rightarrow$ collusive (8 types)
 pure leadership $\rightarrow$ no formal agreement among oligopoly

2) ~~cartel~~ collusive $\rightarrow$ not collaboration
 or non-collusive $\rightarrow$ cartel

* oligopoly $\rightarrow$ non-collusive $\rightarrow$ cartel
 $\rightarrow$ collusive $\rightarrow$ price leadership.
 on that type of market the cartel is a condition where formal agreement / collaboration among the farm exist. (formal agreement)

price leadership $\rightarrow$
 an order to exist in the market, to have market situation, one has to surrender to other farms, one leads others in price.

* barometre $\rightarrow$ what is the bigger farm, to overcome more, already existed one ~~other~~ older farm

duopoly
 part of oligopoly, where 2 farms exist in the market.

dt: 17 Dec 2020

Externalities

* externality occur when producing or consuming any product ~~cause~~ cause an impact on third party not directly related to the transaction.

* it can either be (+) or (-)ve.
 * it can occur through production as well as consumption

* $\rightarrow$ if transaction happen
 $\rightarrow$ will get an impact (may be production or consumption)

* plays very imp role in economics. more eg driving into a city, it will cause pollution in that city.

* it is the consequences of one person's work ~~has~~ that will be his bearred by other.

positive externality is eg $\rightarrow$
 suppose we plant a tree for consumption of fruits, other person will also get that fruit, they can enjoy while sitting in the shadow of the tree.

* a farmer grows apple tree. external benefit is that he produces nutrient for nearby bee-keeper who gains increasing money / more money.
~~benefit~~ money / more money. community bears the cost of individual benefits, socially community bears the cost private cost & social cost

private cost $\rightarrow$ social cost (the externality)

negative externality

* when we consume a good, it cause a (-) impact on 3rd party.

* using cars, we get luxurious travels but

due to pollution it has (-ve) impact on others.

Public Goods And Externalities

Public Good

- * A commodity or service that is provided without profit to all members of a society either by govt., by private indiv. or an organisation.
- * Public goods are those where one individual does not reduce its availability or others exclude others from consuming.

3 types of good

public goods
private goods
merit goods

public goods

- * ~~non-excludable~~ freely & plentifully available
- * ~~conspicuous~~ social responsibility
- * may be given by organisation
- * we don't have to pay.
- public - people good $\rightarrow$ product.

public good

- * those goods which are provided by govt. or organisation which we can enjoy without paying anything.
- * ex: child education, free hospital facility, govt. school.

air, oxygen, culture, alarm, ex: gun, food, bridges

characteristics

- * non-excludable $\rightarrow$ we can not ^{re}debar it.
- * non-excludable person from using it.
- * indivisible $\rightarrow$ we can not divide.
- * non-profitable

- * free riding problem - anyone can enjoy it freely without paying.
- * due to it is not divided, ~~it~~ as enjoyed by all, we get negative externalities.

* non-rivalry

- * all person or a nation ^{can} consume equal.
- * one individual does not exclude its consumption by others.
- * consumption of non-rival goods by an individual does not reduce its amount available for others to consume.
- * ex: enjoyment provided by parks, national defence, television signal, light house.

non-excludability

- * The non-exclusive nature of public good implies that it is difficult if not possible to exclude those from consuming them who are not willing to pay for them.
- * It is not possible or it is very costly to prevent those people who do not pay for these goods.

- * Ex: national defence is a public good and provided to all member and benefit are available to all equally irrespective of whether some people pay taxes for it or not.

Free-Rider Problem

- * Free of cost
- * Non excludability of public good could lead to market failure that is failure of market to achieve pareto efficiency.
- * Producers are unable to prevent from consuming who do not pay their share of cost, there is a problem of free rider arises.

- * so, profit maximizing firm will either not produce it or produce too little.

Efficiency And Public Good

- * Efficiency is achieved when marginal benefit and marginal cost are equal.
- * How much each person values and additional unit of output.

Marginal benefit is obtained by adding this value for all people who enjoy the good.

- * To determine the efficient level of provision we must equate sum of these marginal benefit to marginal cost of production.

Indivisible

- * producers can not divide public good among the people who consume it for free or who pays for it.

Merit goods

- * R.A. measure given the definition of merit good.
- * some positive externalities
- * either free or lower cost
- * eg - education

Demerit goods

- * negative externalities
- * eg - alcohol, tobacco

28 dec 2020

- * the effect of economic activity of two parties on the third party who is not involved in that. $\rightarrow$ externalities.

* Public goods $\rightarrow$ welfare motivated.

freely available

- $\rightarrow$ non-rival in nature i.e. if one person consumes it, it will not affect other,

- eg $\rightarrow$ army force / defense

* Rivalry $\rightarrow$ rival goods (sweets)

- $\rightarrow$ free gift of nature
- $\rightarrow$ non-excludable - does not exclude other person from consuming the product.

from consuming park is non-rival but paying + recreation park is ~~cost~~ rival.

entering into it can be ~~cost~~ rival.

- $\rightarrow$ zero marginal cost of public goods $\rightarrow$

- $\rightarrow$ efficient provision of public goods $\rightarrow$ MRS = MC

these are welfare motivated, so we

everyone can vote freely without paying for it $\rightarrow$ free rider problem.

- $\rightarrow$ in case of public goods there is market failure.

when marginal cost of production is zero it is said to be non-rivalry

- * ~~ex~~ impure $\rightarrow$ public good that gradually loses one or more characteristics of public good.

defense + we all are getting it equally

* defense + we all are getting it equally

so, public good.

- * merit good $\rightarrow$ positive externalities, eg hospital, education, they are free of cost. It can be provided by govt.

→ GII & Gross Happiness Index - given by living standard of people, ~~not~~ Life span, living cost, education etc.
→ Gross Happiness Index.

* General product:-

a product available in a brand is more preferred than the same quality of product available in a roadside.

Marginal cost is equal to zero:-

* free rider problem leads to market failure, ~~cost~~ to achieve Pareto efficiency, MC of admitting additional person to consume the good is zero, then Pareto efficiency requires the price of public goods should be zero.

→ But total production is not zero to produce, public good is very expensive. To meet this, private marginal cost is higher than Pareto efficiency concern and less than market failure to this leads from market failure to Pareto efficiency.

24 Feb 2022

Inflation

defn

* change in value of money associated with the change in general price level, & forms to describe these changes.

→ deflation } associated with price level fallation

price level

when ~~the~~ average price of all the goods & services available in the market.

Inflation - situation when there is continuous rise in price of the product.
→ unjust

deflation - situation when there is continuous fall in price of the product
→ very bad / worst for economy

type of inflation

- ① creeping / very mild inflation (0-3%)
- ② walking / creeping inflation (3-7%) or less than 10%
- ③ galloping inflation (good for economy)

price rise within 10-16%

hyperinflation

more than 16%

the causes of inflation (two)

- ① demand pull
- ② cost push

→ demand pull inflation

when there is excess demand, compared to the const. supply, price increases factors that are responsible

money supply

→ if money supply increases, price increases, demand increases

10 product → 1000s - 1st cost - 1000s

10 product → 1000s + 100

cost increases, 1100s

→ disposable income

α disposable income, price ↑

→ increase in business expenditure

α demand, price ↑
people get employment, will have more disposable income

↑ increased foreign demand :-
↑ ~~at~~

Cost push inflation theory

~~scarcity~~ demand remains constant, due to the cost or factors of production supply decreases. ∴ price increases.
Factors responsible

① ~~increase~~ wages / price
if ~~a~~ wedges ↑, supply ↓, price ↑

② higher taxes

more taxes, supply ↓, price ↑

③ fall in the availability of basic input

④ * Major tax taken by Govt. to control
inflation &
stages

⑤ Ray RAI

↑ generally RAI increases rate at
time of inflation & decrease rate
at time of ~~recession~~ deflation.

⑥ Ray Govt
↑ imposition of more tax on govt expenditure,

monetary policy

Ray RAI
more rate of interest
↑ control money
supply.

RAI

↑ Apex Bank
↑ controller of monetary system.
April 1, 1935, established, nationalised in 1948

funcⁿ

① Acts as clearing house of country
clear alligates of bank
clear alligates of bank
clear alligates of bank

② financial adviser of ~~the~~ govt, agent.

③ Banker's Bank
wherever necessities RAI ~~has~~ lend to
wherever other bank.

④ Banker to govt

⑤ controller of credit & foreign exchange

⑥ custodian of foreign value of currency

⑦ ensure the internal value of currency

⑧ fight against economic crises & stabilises

price stability

⑨ guardian of large resources.

⑩ manager of note issued, all notes are
printed by RAI

⑪ publisher of financial & economic
statistics.

⑫ credit to weaker section, credit guarantee
moneylending & other data are published
by RAI.

Commercial bank

ex RAI, funcⁿ
commercial bank
inflation, type, ~~the~~ measure to control
overview of Indian bank.

Dt- 22/sept 2020 > Pg 145

M-4/pg 98

economics → financial management of household.

economics comes from ^{greek} word, ~~eco~~

eco & nomos
↓ ↓
home accounts.

In business give & take relations.

→ factors of ~~the~~ marketing (7 factors) →

man, material, machine, money, management

→ four type of Economic utility:-

1) place

2) price

3) promotion

4) time

Management

management → man of any age who is able to get max^m satisfaction using limited resources available.

Management (defn)

It is the art of getting things done through others & having them do it willingly.

Processes in selection of a project :-

Break-even analysis

1. establish relationship b/w rev, revenue & profit.

cost → the amount of resources that are sacrificed to get or procure a particular thing.

Revenue → the amount you get while selling a product

Profit = Revenue - Cost.

Elements of cost :-

- 1) material → raw material
- 2) labour / wage
- 3) expenses

→ direct material
the material which are directly identified with a finished product.

eg:-

for making shoe
leather, timber is
required for furniture.

→ indirect material

indirect materials cost are negligible but

they are required to make raw material to finished product
eg: nail

labour or
words those are engaged to convert the raw material to finished product.

① Direct labour → the directly engaged workers

eg: carpenters, welders, machinist wages

② Indirect labour → Not directly engaged.
eg: sweeper, watchman, store keeper wages.

expenses

direct expenses → those expenses which are directly allocated for particular job

eg: for welding work electricity is

direct expense.

indirect expenses → not directly allocated

expenses.

eg: for classroom teaching electricity

is indirect

Prime cost = direct material + direct labour / wages + direct expense

Overhead

the sum total of indirect material, indirect labour & indirect expense.

③

* Prime cost is also known as marginal cost, that cost, first cost, direct cost.

Overhead type (A/c to nature)

→ factory

→ office

→ selling

→ distributing

Research & development.

① factory overheads include all the expenses made on the actual operation of producing a product like indirect material & indirect labour eg → factory rent & lighting, insurance etc.

② office overheads:- ~~these~~ ^{these} are all indirect expenses, ~~labour, material~~ ^{labour, material} when formulating a plan for office, say even head →

All the indirect expenses, expenditure, labour incur in the selling department eg. advertisement, sales manager, sales representative, agents, ~~salesman~~ ^{salesman} salaries, cost of publicity.

③ distribution overhead → delivery van charges, packing material. sometimes cost of packing is more than cost of material/product.

④ Research & development overhead → It is based on system approach & most imp type of overhead. eg → cost of raw material used in research, staff cost involved in research.

defn All the indirect materials involved in forming a product are called overhead cost and

A/C to variability overhead cost are

- ① fixed
- ② semi-variable
- ③ variable.

fixed cost

are costs which remain const. irrespective of volume of product.

we can not control the fixed cost. Total fixed cost remains const but

due to increase in volume in production the cost per volume ~~decreases~~ ^{decreases}.

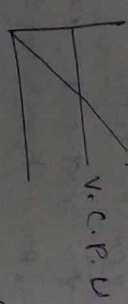


variable cost

the cost which ^{proportionately} increase in volume of production.

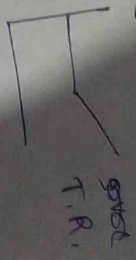
* total variable cost increases but variable cost / unit volume production remains const.

* It can be controlled



⑤ semi-variable cost the cost which remain const. upto certain point & then increase.

eg → telephone rate.



upto 5 calls - const. then per unit some Rs.

Break-even analysis is a part of marginal cost which helps the manager to analyze the financial part & take decision.

main object of B.E.A. $\rightarrow$ to calculate how much unit we have to produce so that total cost = total revenue.

B.E.A. is based on the concept of production. If production will increase the

2 Approach

1) Linear $\rightarrow$ volume of production & cost
2) non linear

Assumption in linear approach,

1) selling price & remain cost in competition

2) variable cost remain const.

3) total fixed cost remain const

4) whatever produced & sold out, no unsold or closing stock, all units produced are sold.

5) algebraic method 6) graphical method

ratio b/w construction per

up marginal safety

dt: 29/11/2020

B.E.A. establishes relation between cost, revenue & profit. It is part of marginal cost & variable cost.

Marginal cost is any increase in total cost due to addition of unit of production.

cost & revenue are a function of production. cost & revenue $\rightarrow$ less cost $\rightarrow$ max profit.

more produce $\rightarrow$ less cost $\rightarrow$ max profit.

Break even point :- the point of no profit, no loss, total cost = total sell

variable cost & ~~production~~ revenue

linear approach

direct / proportional relationship.

variable cost.

fixed cost

Break even point

units of production

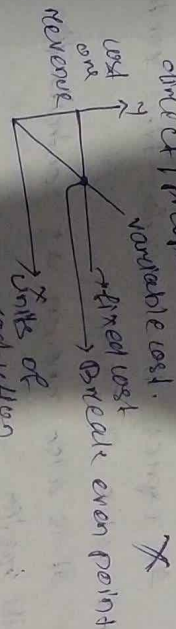
Non-Linear Approach

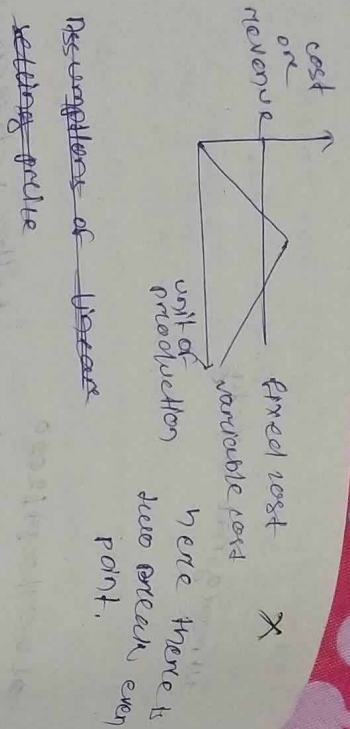
product life cycle $\rightarrow$ stages

infant, growth

saturation

declining





Contribution → Excess of sell over variable cost
 Mathematically / symbolically :-

$$C = S - V \quad P = \text{Profit}$$

$$L = 1000$$

$$C = F \pm R/L \quad F = \text{Fixed cost}$$

contribution per unit =

SP per unit - variable cost / unit
 + selling price / percent remain constant.

Profit volume ratio

It express the relation b/w contribution & value of sales.

It is also known as contribution to sales ratio.

It is usually expressed as a % & helps the manager to guide the profitability.

$$P/V \text{ ratio} = \frac{\text{contribution}}{\text{sales}} \times 100$$

$$P/V \text{ ratio} = \frac{\text{change in profit}}{\text{change in sales}} \times 100$$

$$P/V \text{ ratio} = \frac{\text{fixed cost}}{\text{break even sales}} \times 100$$

↑ higher the P/V ratio better the financial position.

③ Break even point
 The point of neither profit nor loss, or total cost = total sell.

$$B.E.P = \frac{\text{fixed cost}}{\text{contribution per unit}} \quad (\text{unit})$$

$$B.E.P \text{ sales} = \frac{\text{fixed cost}}{P/V \text{ ratio}} \quad \frac{F + \text{desired profit}}{P/V \text{ ratio}}$$

B.E.P. sales = to earn a desired profit.

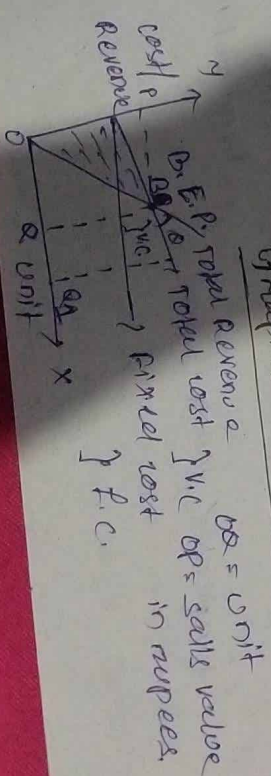
④ Margin of safety :-
 difference b/w Actual sell & Break even sales.

$$A.S - B.E.S = M.S. \text{ in } \text{Rs.}$$

$$M.S = \frac{\text{profit}}{P/V \text{ ratio}} \quad \text{Actual sell} - B.E.S \times 100$$

$$M.S \text{ in } \% = \frac{A.S}{\text{Actual sell}} \times 100$$

Graphical Method



It is angle of incidence dotted area - loss curve.

$DB_1 - DB_2 = DB_1$ = margin of safety.

B.E.A. in linear approach.

total cost $[DB_1 + V.C. + I.C.]$

At: 2 October 2020

example

Depreciation

It is derived from within word
de. pre. tion.
↓
decline price

Depreciation means decrease in the value of ~~property~~ ^{fixed} asset except loss due to wear & tear & continuous use.
two type of asset:

Asset means the properties or business having money value.

It is of two types: 1) current asset
2) current asset: the benefit is derived from the asset in same year (for immediate consumption)

3) fixed asset: those assets the benefits of use are going to derive throughout life.

~~due to wear & tear~~ +

* current assets are not depreciable.
cause of wear & tear - fire, theft
+ chemical reaction

tangible (touch & see)

Types of depreciation

Accounting basis - (due to passing of time).

→ economic basis (due to change in market value)
→ physical changes basis (rain, water, dust)
→ change in physical value (sunlight, accident)

economic basis → change in market economy
due to passage of time.

Accounting basis: At end of every year the book value is reduced by a fixed amount, due to passage of time.

Depreciation
It is a permanent, continuing & gradual shrinking of book value of fixed asset.

It is appropriation to profit, not charge against profit.

* change against profit eg rent of house.
depreciation is a loss as well as expenses, depending upon its use (personal / business)
→ eg washing machine 20,000
10 yrs

DP - 2000.

loss →

→ If wash machine buys washing machine for business purpose, then it is expenses because, it will generate some profit.

Feature of depreciation

It is the part of asset which is consumed to generate some revenue.

→ All tangible asset except land depreciate.

→ Fall in the value of such assets must be continuous & gradual (fixed). It is example of loss as well as expenses.

→ maintenance of assets, does not mean the value is not depreciated.

→ At the end of ^{estimated} life of particular asset if we dispose the amount we used that value is called scrap value. It is an income.

→ Book value is:
At what value the thing is purchased.

$$B.V._1 = 10,000$$

$$\text{depreciation} = 1,000$$

$$B.V._2 = 9,000$$

$$B.V._2 = \text{Book value at end of 1st yr}$$

$$B.V._3 = 8000$$

$$\text{cycle time} = 1000 \text{ km}$$

$$\frac{200}{1000} = 0.20 \text{ rs.}$$

$$10 \text{ km} - 10 \times 0.20$$

at the end of 1000km, it is not to be further used.

$$1000 \times 0.20 = 200 \text{ rs.}$$

* e/c to Indian accounting standards, (GAP) generally accepted accounting principle is needless any its to spread the value of asset in lifetime life time,

so that we can derive the actual amount.

need for depreciation:

→ To know the net profit/loss

→ To know the value of asset.

→ To know the replacement cost of asset.

→ Real value - (no. of yr $\times$ depreciation)
if in this case we change depreciation 10%, the other party is going to change 20%.

* due to maintenance & productivity will increase.

concept of conservatism = irrespective of profits a certain amount of money must be kept for depreciation.

* ~~non~~ human being are also assets.
cause of depreciation:

→ wear & tear (friction, ~~the~~ vibration intensity, chemical reaction)

→ due to aging (quality & ability declines)

→ depletion (assets like oil well, coal mine, stone lost their value due to extraction of minerals from them. Loss of or wasting assets due to extraction of minerals)

→ accident

→ incapacity

→ natural causes (eg. electric wires when exposed to sunlight, insulated wires

is damaged.

→ on pens due to salty weather, they material rusting

→ obsolescence

→ due to research & development ^{continuous} of ~~new~~ ^{newer} technological asset, we replace or disposing old ones at a lesser price.

→ effect of ~~the~~ economic loss

* suppose a land is taken at 10,000 in for 10 yrs at lease from govt., at depreciation 1,000. After 10 yrs it should be renewed.
+ also copyright.

→ renewal
if renewal is given to add engine oil, when don't follow it, it would damage.

At - 6 Oct 2020

Methods of computing depreciation:-

there are no. of methods for calculating depreciation but mainly following 2 methods are used (include in syllabus)

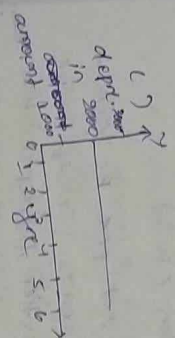
the 2 methods which are most imp. are:-
1) straightline method / fixed installment method,
2) original cost method

3) diminishing balance method / written down value method / Reducing balance method.

straightline method:-

It is the simplest method to apply & ~~also~~ most widely used.

under this method, the amount of depreciation remains same for each yr till the end of the economic life of asset.
As the annual depreciation remains const., if we plot on a graph upon it will produce a straight line.



→ the book value of the asset is equal to diff. b/w the purchase price & scrap value of the asset.

→ depreciation under straightline method can be calculated by applying the following formula:-

$$D.C. (CN) = \frac{P - S}{N}$$

1) $D.C. (CN) = \frac{P - S}{N}$
on original cost - scrap value

estimated life in years

P → original cost / purchase price of asset

S → scrap value / salvage value

N → estimated life in years

D.C. → Depreciation charge

2) Rate of depreciation (% of depreciation)

$$d = \frac{D.C.}{P} \times 100$$

3)

$$BV(n) = P - n(D.C.)$$

or

$$BV(n) = P - n \left[\frac{P-S}{N} \right]$$

$n \rightarrow$ period

Even $n \rightarrow$ B.V. at end of n yrs.

n = depreciation calculating yrs

BV = Book value

d = rate of depreciation (in %)

eg. A machine is purchased for Rs 7000 & the life period is 5 years. Scrap value

nil. Calculate (1) depreciation, (2) depreciation reserve

and (3) B.V. of asset at end of 3rd yr.

Final the rate of depreciation.

$P = 7,000$

$N = 5$

$S = 0$

$\therefore D.C. = \frac{P-S}{N} = \frac{7000-0}{5} = 1400$

1) Depreciation reserve = $3 \times 1400 = 4200$

2) B.V. = $7000 - 3(1400)$

B.V. = 2800

~~B.V. = 413.94~~

3) $d = \frac{D.C.}{P} \times 100$

$= \frac{1400}{7000} \times 100$

$= 20\%$

* calculate depreciation from date of purchase to end of accounting period.

suppose 1st year - 31st Dec, 10% per annum use start 1st Jan 1st, 10,000 (rate of depreciation)

$10,000 \times \frac{10}{100} \times \frac{7}{12} = 700$

* if the date of purchase is not given, use go for 6 months

$10,000 \times \frac{10}{100} \times \frac{6}{12} = 500$

* if rate of depreciation is 10%, if per annum is not given, we have to calculate for 1 yr.

Merit of straight line method:-

is simple to calculate & easy to implement

if there is less record to maintain &

time minimisation

it is suitable for those assets having fixed life i.e., building, patent, furniture etc.

up the value of asset become zero at the end of its estimated life.

Demerit:-

The method of calculating depreciation is illegal because an asset is more

productive initially than in later years, therefore amount of benefit & depreciation are not matched properly.

* This method is not recognised by ^{the} taxation

Income Tax Act 1961

* Due to passage of time repair charges increases but depreciation remains const.

* It ignores opportunity cost of investment
opportunity cost → the best alternative price.

Diminishing balance method

→ It is also known as written down value method or reducing balance method.
→ This method is based on the assumption that an asset is more productive in the initial years than in the later years, as its mechanical efficiency declines with passage of time, hence larger amount of depreciation should be charge in the initial years & smaller amount of depreciation later on.

→ Under this method depreciation is charged at a fixed % on the diminishing balance every year.

ex:- Asset value 10,000
Rate of depreciation 10% p.a.

St. method	Diminishing balance
10,000	10,000
100	- 10%

formula of diminishing balance method:-

$$D(C_n) = \frac{R}{n} \text{ even } (n-1)$$

R = Rate of depreciation

$$R = 1 - \left[\frac{S}{P} \right]^{1/n}$$

P = purchase price

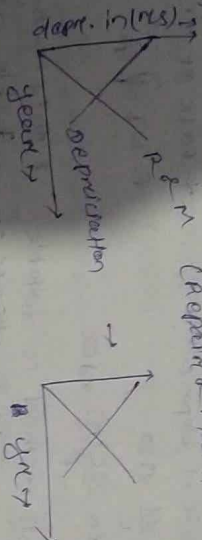
S = scrap value

n = estimated life in yrs

R = Rate of depreciation

BV = Book value

$BV(n) = P(1-R)^n$ (Repair & maintenance)



Merit

- 1) There is fair spread of depreciation & repairs
- 2) It helps management in paying less tax initially
- 3) It reduces the effect of obsolescence.
- 4) It is recognised by Income Tax Act.
- 5) It is suitable for assets where repair plays imp role.

Demerit

- 1) Asset can not be completely written off.
- 2) It is difficult to calculate the rate of depreciation.

Diff b/w straight line & diminishing value method:-

- | | |
|--|--|
| <p><u>straight line method</u></p> <ul style="list-style-type: none"> * depreciation is calculated on the original cost. * Amount of depreciation remains same on same * Value of asset reduces to zero after the expiry of economic life. * Rate of depreciation $R = \frac{C}{P} \times 100$ on $\frac{D(CN)}{P} \times 100$ * This method is suitable for repair & change of obsolescence is less * Net profit is not affected. | <p><u>diminishing value method</u></p> <ul style="list-style-type: none"> * depreciation is calculated on the book value. * Amount of depreciation gradually reduces * Value of asset is never reduced to 0, after the end of its economic life. * Rate of depreciation $R = 1 - \sqrt[n]{\frac{S}{P}}$ * This method is suitable where repair goes on increasing & obsolescence is high * Net profit is not much affected. |
|--|--|

Time Value of Money

Interest plays a key role in determining it.

Value of money is decided by:-

- 1) As can be invested & earned by interest
- 2) It is potential earning

1) money due to ~~invested~~ inflation → earning away? the purchasing power of money with is falling away by inflation.

2) ~~investing~~ (in money) at the time the earning power of money increases and purchasing power is gradually decreasing away because of inflation.

~~10000~~ 20000/- rate - 5%
 10000 20000/- → 5% rate - 10%
 in 10000 35000/- → 2% rate - 10%
 10000's money is better than 10000's money.

Things we must consider to collecting the value of money:
 time rate present value future value

CPI → cost inflation index

Interest :- It is the monetary benefit the owner of money will get by lending money to others.
 It is the fee paid with borrowed money / compensation in shape of money to the lender.

rate of the interest is higher in case of private moneylender than bank because the private is more risk in case of private moneylender.

Simple interest is the amount of interest which is calculated on the basis of principles irrespective of principal.

Principal = 10,000
rate = 10%
time = 3 yrs

$$S.I. = \frac{P \times R \times T}{100} = \frac{10,000 \times 10 \times 3}{100} = 3000$$

Future value = 13,000

Amount / future value = $P + SI$

$$\text{Principal} = \frac{\text{Amount} \times 100}{R \times T}$$

1 yr = 24 fortnights

Compound interest

The amount of interest which is calculated on the accumulated balance of previous yrs.

There are 2 methods of calculating C.I. →

- ① gradual method
- ② Algebraic method

gradual method

P = 1000
rate = 10% / annum

t = 3 yrs

$$1st \text{ yr} = \frac{1000 \times 10 \times 1}{100} = 100$$

$$F.N / A = 1000 + 100 = 1100$$

$$2nd \text{ yr} = \text{Interest} = \frac{1100 \times 10}{100} = 110$$

$$F.V. = 1100 + 110 = 1210$$

$$3rd \text{ yr} = \text{Interest} = \frac{1210 \times 10}{100} = 121$$

$$F.V. = 1210 + 121 = 1331$$

S.I.

$$\frac{1000 \times 3 \times 10}{100} = 300$$

$$F.V. = 1300$$

Algebraic method

$$F = P \left(1 + \frac{R}{100} \right)^n$$

$$C.I. = F - P$$

F → future value

P → present value

R → rate of interest

n → time

$$F = P \left(1 + \frac{R}{100} \right)^n$$

$$C.I. = [P \left(1 + \frac{R}{100} \right)^n - P]$$

→ Half yearly

$$F = P(1 + i/2)^{2n}$$

rate %
time x 2

→ Quarterly

$$F = (P + i/4)^{4n}$$

→ monthly

$$F = (P + i/12)^{12n}$$

→ Continuously

$$F = (P + i/24)^{24n}$$

If ~~time~~ is not in exact number i.e.,

$2\frac{1}{3}$ yrs

$$F = P(1 + i)^n (1 + ni)$$

$$P = 1000$$

$$R = 10\%$$

$$T = 2\frac{1}{3} \text{ yrs}$$

$$F = 1000(1 + 0.1)^2(1 + 0.05)^{\frac{1}{3}}$$

$$(1 + 0.1)^2 \times (1 + 0.05)^{\frac{1}{3}}$$

Ex:- 16 Oct 2020

Normal rate of interest:-

when rate of interest is calculated on annual basis. (10% per annum)

Effective rate of interest:-

when rate of interest is given monthly quarterly & v have to calculate / convert

100 10% per annum. for 1 yr

$$100(1.1) = 110 - 1\text{yr}$$

$$11 = 10 - 100 \times 10\%$$

$$100(1.05)^2 = 110.25$$

$$10.25 = 10 - 100 \times 10.25\%$$

if into yearly rate of interest, then it is called effective rate of interest.

$$\text{eff}(i) = \frac{F - P}{P} \times 100$$

$$\frac{110.25 - 100}{100} \times 100$$

$$= 10.25$$

$$A = P(1 + i/4)^{4n}$$

$$A = P(1 + i/365)^{365n}$$

$$P = \frac{A}{(1 + i)^n} \text{ or } A(1 + i)^{-n}$$

when A & i are given

$$A = 1331$$

$$R = 10\%$$

$$T = 3 \text{ yrs}$$

$$P = A(1 + i)^{-n}$$

$$= 1331(1.1)^{-3} = 1000$$

→ to find rate when amount, principal & time given.

$$A = P(1 + i)^n$$

$$A/P = (1 + i)^n$$

$$\sqrt[n]{A/P} \approx (1+i)$$

$$i = \sqrt[n]{A/P} - 1$$

$$n = i \times 100$$

to find out time when amount, rate & principal given:-

$$A = P(1+i)^n$$

$$A/P = (1+i)^n$$

$$\log(A/P) \leq \log(1+i)^n$$

$$\log A - \log P = n \log(1+i)$$

$$n = \frac{\log A - \log P}{\log(1+i)}$$

$$\log 0.000045 = \underline{\hspace{2cm}}$$

characteristic matches can be (-, +, 0)

in betⁿ decimal point & 1st significant digit no. of zeros = 4

$$- (4+1) = \underline{-5}$$

Father of logarithm $\rightarrow$ John Napier

common - base 10
natural - base e

$\log$ + another 2 number.

characteristic matches

characteristic of 2234 $\rightarrow$ 3

if no. is > 1

characteristic = no. of digit in no - 1

$$\text{here, } 4 - 1 = 3$$

characteristic (223.4)

here 34 is 3 digit no.

$$\text{so, } 3 - 1 = 2.$$

characteristic of 12.34

$$\rightarrow 2 - 1 = 1$$

12 $\rightarrow$ integral part
34 $\rightarrow$ decimal part

characteristic of

$$0.1234 = \underline{-1} - (0+1) = \underline{-1}$$

characteristic of

$$0.01234 = - (1+1) = \underline{-2}$$

characteristic of

$$(5)^{46}$$

$$\log b_a = \log b_c \times \log c_a$$

$$\log 9 = 1$$

$$\log 10^4 = 0$$

$$9^3 = 8 \rightarrow \log 9 = 3 - \text{real value.}$$

base

$$\log(12+2+3) = \log 1 + \log 2 + \log 3 = 1 + \log 6$$

$$\log(4+3) = \log 4 + \log 3$$

doubling

at particular n , after how many yr the amount is doubled, under compound interest :-

2 formulae

$$\textcircled{1} \text{ Rule } \frac{72}{r} = \frac{72}{\% \text{ or } n}$$

$$\text{if } r = 10\% \Rightarrow n = \frac{72}{10} = 7.2 \text{ yrs}$$

100 become 200 after 7.2 yrs

$$200 = 100(1.1)^n \Rightarrow 2 = (1.1)^n$$

$$n = 7.2$$

② Rule 69

It is more appropriate & gives more accurate result

$$n = .35 + \frac{69}{\text{rate of interest}}$$

$$= .35 + \frac{69}{10} = 7.25 \text{ yrs.}$$

annuities

* the rate of goods is 40,000, then if the buyer is unable to pay, then there is a agreement stating 10,000 down payment & rest 30,000 in 6 installment.

* It is derived from cash world money, which means periodic

payment of equal amount to be paid to meet ones obligation.

It is based on equal payment series. compound interest $\rightarrow$ single payment series.

$$\text{B. suppose } 12,000 \text{ in 3 yrs installment.}$$

$$4000 + 12000 = 5200 - 124$$

$$3000 - 4000 + 800 = 4800 - 2nd$$

$$4000 + 400 = 4400 - 3rd$$

total 14,400 payment.

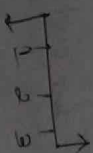
4000 - principal.

this highest purchase system

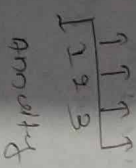
annuity :- 12,000 in 3 yrs, each yr 5000.

at 5/11/2020

Annuity



C.I



annuity.

types of annuity

① annuity certain

120,000 - down payment

4,00,000 $\rightarrow$ in 10 installment

each - 40,000 + interest

total 500000

here no. of installment is mentioned.

③ Annuity perpetuity ^{forever}

no. of installment is not given, it is continued throughout the lifespan.

eg: Nobel prize

sm Alfred Nobel → dysphantic discover.

we have to make the payment forever.

suppose rate of interest is 10%.

the total amount = 10000

so, every yr $\frac{10000}{100} \rightarrow 1000$ have to be paid.

④ Annuity deferred (delay)

total amount = 500,000, 10% rate.
at the end of 3rd yr installment begins.

⑤ Annuity immediate

all annuities are paid at the end of the yr.

* All annuities except deferred are immediate annuities.

⑥ Annuity prepaid

* The annuity is paid at the beginning of yr.

eg: all type of insurance policy premium

life insurance premium

PPC → general insurance (property insurance)

LIC → life insurance

⑦ Annuity contingent

the happening of event is not known,

suppose Mr X takes 100,000 from Mr Y.

saying that he will return him 200,000

as when he take over the business

business from his father, after his father's death.

suppose father's age = 80.

Mr Y has to wait till his death to get the extra 100,000 money.

eg: study loan.

suppose after 4 yrs back we have to wait 1 yr for job.

here in general after 5 yr study loan is paid, it can also be done after 4 yr.

Cash flow diagram

when we are going to start any kind of project, that project's called as capital expenditure.

it is the benefit we are going to derive from no. of yr → capital expenditure.

two type

cash outflow → cost of the project

cash inflow

benefit (end of each yr on revenue or receipt on receipt)

earning except insurance (always +)

Cash out flow

→ occur at beginning of period.
→ called as disbursement
→ always negative

Three stages cash flow diagram

operational stage
~~operational stage~~

initial stage

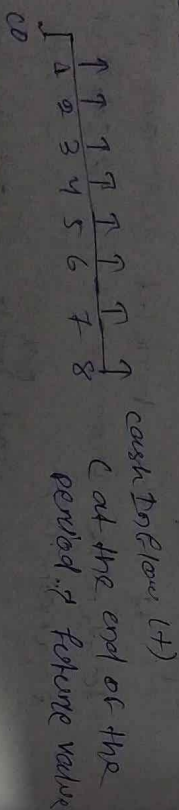
① initial → cash ^{out} flow till the production is started.

② operational → once ~~cash flow~~ production is started till the end of production is dissolved.

③ Terminal → disposal of all assets

Defn

The pictorial representation of cash flow & cash out flow in a time scale is known as cash flow diagram.



cash outflow } money spent at the beginning
present value

Net cash flow

→ cash inflow - cash out flow

→ It is the sum of receipt & disbursement.
(4)

Types of cash flow :-

Initial → once the construction is started till the process ends, the cash outflow always negative (-).

operational → from the day operation is started till the end of production.

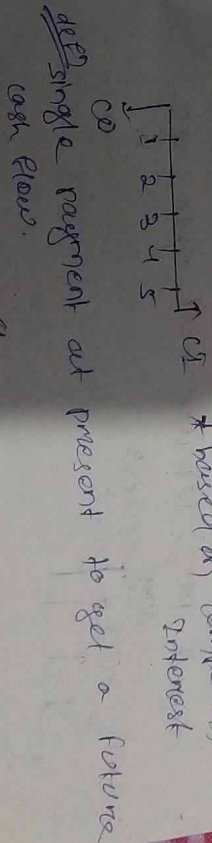
→ both positive & negative i.e., both cash inflow & outflow
→ amount of reinvestment less than amount of benefit derived.

Terminal → disposal of all assets after completion of project
→ positive (only cash inflow)

Types

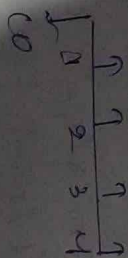
① single cash flow :-

→ only one cash inflow & one cash outflow
→ one time disbursement & one benefit
→ based on compound interest



② equal cash flow

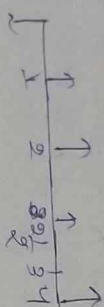
→ no. of benefits are derived.
→ magnitude of various & distance b/w & annuities equal.



Defn periodical payment of equal amount based on annuity

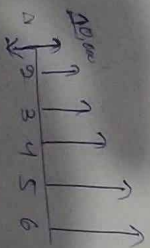
② Unequal / unequal cash flow:-

magnitude & distance
b/w 2 arrows are equal
periodical payment amount & periods are
not equal.



③ Unequal Gradient cash flow:-

In this cash flow there is a fixed amount
of increment every yr.

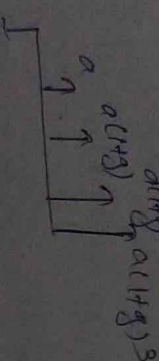


1 → 10,000 $a+g$
2 → 11,000 $a+2g$
3 → 12,000
6 → 15,000 $a+(n-1)g$

3rd gradient

④ Geometric gradient cash flow diagram:-

* There is a fixed rate or % of increment
or decrement
deposit 1000, 10% increment every yr.



diff types of cash flow problems:-

① Single payment compound amount factor:-

To find F, when (P, i, n) given

F → Future value

P → Present value

n → time & i → rate

$$F = P(1+i)^n$$

② single payment present worth factor:-

$$(P/F, i, n)$$

$$P = F(1+i)^{-n}$$

$$P = \frac{F}{(1+i)^n}$$

$$F = 500000$$

$$10\% - i$$

$$P = ?$$

$$10 \rightarrow n$$

$$\text{present value} = \frac{5 \times 10^5}{(1.1)^{10}}$$

③ equal payment series compound amount factor:-

$$\text{objective } (F/A, i, n)$$

A → annuity or annual equal amount

$$F = \frac{A}{i} \{ (1+i)^n - 1 \}$$

$$A \rightarrow 10,0000$$

$$i \rightarrow 10\%$$

$$n \rightarrow 5 \text{ yrs}$$

$$F = ?$$

④ equal payment series sinking fund factor:-

$$\text{objective} \rightarrow (A/F, i, n)$$

sinking fund

A fund which is created to get a
lumpsum amount in future.

$$A = \frac{F \cdot i}{[(1+i)^n - 1]}$$

⑤ equal payment series present worth factor

Objective $\rightarrow (P/A, i, n)$

A annuity
 $i \rightarrow$ Rate
 $n \rightarrow$ time

$$P = \frac{A}{i} \left\{ 1 - (1+i)^{-n} \right\}$$

suppose we deposit 12,00,000 rs at the end of each month 6,000 rs. mths $\rightarrow$ monthly interest scheme 100,000 $\rightarrow$ each month 5000 rs deposit

⑥ equal payment series capital recovery factor

Objective $\rightarrow (A/P, i, n)$

$$A = \frac{Pi}{\left\{ 1 - (1+i)^{-n} \right\}}$$

⑦ Linear gradient series annual equivalent amount factor

Objective $\rightarrow (A/A_1, G, i, n)$

$A_1 \rightarrow$ initial payment
 $G \rightarrow$ fixed amount of increment
 $i \rightarrow$ Rate
 $n \rightarrow$ time
 $A_1 \rightarrow 10,000$
 $G \rightarrow 1000$
 $i = 10\%$

$A \rightarrow$ annual equivalent amount we need to pay

$$A = A_1 + G \left[\frac{(1+i)^n - 1}{i(1+i)^n - 1} \right]$$

⑧ geometric gradient series

Objective $\rightarrow (P/A_1, g, i)$

$A_1 \rightarrow$ initial payment
 $g \rightarrow$ fixed Rate of increment
 $i \rightarrow$ Rate
 $n \rightarrow$ time

$$P = \frac{A_1}{i-g} \left\{ 1 - \frac{(1+g)^n - (1+i)^n}{1+g} \right\}$$

$i \neq g$, this formula is used.
 when $i = g$, $P = \frac{A_1 n}{1+i}$

Principle of Economic Equivalence

Economic Equivalence

* same concept in time value money

* helps in comparison of projects (2 or more)
 i.e., economically

two things are said to be equivalent when they have the same effect
 same effect
 same mechanism

an economic is used, 2 to compare the project involved, which one is eventually producing more profit

require $\neq$ economic apply

two things are economic equivalent when they produce same effect, same advantage

Principle / Rules of Economic Equivalence

① ~~Economic~~ there is equivalence between two cash flows

1000rs → procedure → 110 (after 1 yr) made 10% / Ann

$$P = 1000 \quad F = P(1+i)^n$$

$$1 = 1000(1.1)^1 = 1100$$

Reverse case $(P/F, i, n)$

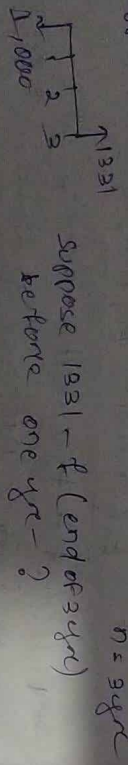
$$P = \frac{110}{1.1} = 1000$$

When present value is given & finding future value & vice-versa, both have the same effect, same amount.

* Financial, economically technical concept of two products same

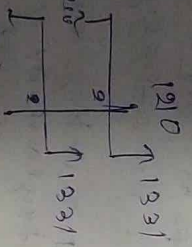
② Economic equivalence analyses at a common point of time.

Suppose 1,000rs → becomes → 1,331 10% / Ann



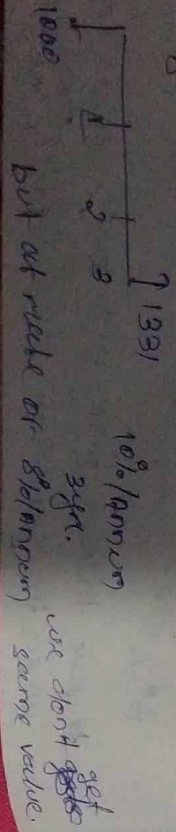
$$P = \frac{1331}{1.1^3} = 1,000$$

2- 1,000(1.1)^3 = 1,331



When, Present value is given 1000 & we have to find future value & vice-versa, both economic equivalence will be same at a point of time

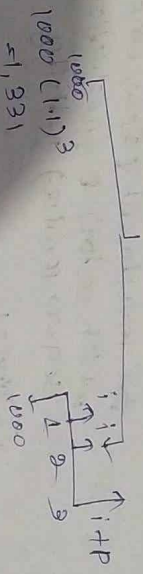
③ Economic equivalence is destroyed due to change in rate of interest.



* Economic equivalence is based on constant in rate of interest.

④ Economic equivalence analysis regardless of point of view of borrower & lender.

eg → we want to take 1,000rs loan at i = 10% for 3 yrs. (options)



⑤ eg → LDC policy of retirement benefit (rate of interest const.)

* → Regardless of point of view there is economic equivalence.

Rate of interest

10% interest, rent 900/month

10% interest, rent 105 → economic equivalence

but if rent is 100, no economic equivalence

rate of income > rate of interest → economic equivalence

Evaluation of Engineering Project

capital budgeting project: completion of project

requires more than one yr. (2 types)

① fixed capital → one time investment

amount invested for & ② working capital.

fixed assets. eg → but at 100, 10% of bank care on rented

basis. construction cost suppose 50 lakh, per month 30,000 rent, so, they pay

1 lakh 40,000 per yr which is less than 5 lakh so, investment is

wise. we have to utilise our capital

* we have to increasing assets & get more resources for increasing assets & benefit.

A project is an activity in which we spend money. Life is also a project, we invest to get some benefit.

Project have (1) starting point & ending point
(2) cost & ~~benefit~~ ^{benefit} ~~return~~ ^{return}
(3) well defined time period for investment (how, when, which sector) & production (sequence wise) activity.

Method of evaluating:-

A project is an imp activity where we expend capital resources to producing asset which we can ~~use~~ ^{expect} to derive benefit over an extended period of life. A project itself includes logically & planning, financing & implementation of resources.

A project has the following feature:-

- (1) It should have starting point & ending point.
- (2) Its cost & returns are ~~measurable~~ ^{measurable} (in terms of money)
- (3) The time sequence of investment & production activities are well defined.

Methods of evaluating

- (1) Payback period method
- (2) Net present value method
- (3) Net future worth method also future value method
- (4) Net present equivalent method
- (5) Internal rate of return method

1. Payback — 25000 payback period
payback period = $\frac{1 \text{ lakh}}{25000} = 5 \text{ yrs.}$
cash inflow

so, the project have small pay back period is profitable.

post-profitable payback period amount is 1 lakh. Cost 5yr we get another 1 lakh

stimulating cash flow

20,000	cash flow in	25,000	1
50,000		30,000	2
90,000		40,000	3
130,000		50,000	4
180,000			

5yr investment (1 lakh)

25,000 return.

when comparing we consider only present value.

investment made by a person in present.

or capital venture.

Dr. 20 Nov. 2020
The financial analyst will take the decision ~~to~~ ^{on} the basis of feasibility of project.

1. Present worth method.

(1) Present worth method are taken into factors & time elements ^{that of} into consideration money is better than tomorrow's money. It is based on this concept.

cash out flow - immediate cash inflow + end of period cash inflow of all the benefit ~~at the~~ ^{to} present worth of all the benefit in future, we are going

comparison of that with present cash inflow, thus we find whether to ~~have~~ accept or reject that project.

steps of present worth method:-

- ① Estimate the project length (n)
- ② Estimate the rate of return (i)
- ③ Calculate the cash outflow (cost)
- ④ Estimate the cash inflow (benefit)
- ⑤ Estimate the net present value (NPV) or net present worth (NPW) i.e.,

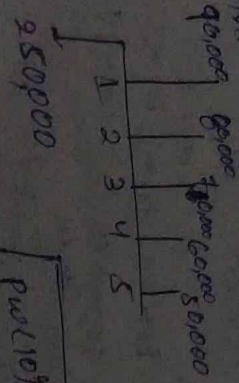
NPV's present value of - present value of cash inflow

and denote it as $PW(i)$ or NPW or NPV

- ③ If $PW(i) > 0$, accept the project
- if $PW(i) < 0$, reject
- if $PW(i) = 0$, remain indifferent

Telecommunication can accept project with $PW(i) = 0$, as it can ~~afford~~ afford losses.

Q) Project A cost rupees ~~250000~~ 250000 & is expected to generate 90,000, 80,000, 70,000, 60,000, 50,000 for 5 yrs. The rate ~~of~~ ^{at} $i = 10\%$, evaluate the project under PW method.



$i = 10\%$
 $n = 5 \text{ yrs}$
 $C.P. = 250,000$

$$PW(10\%) = PW(10\%)_{CI} - PW(10\%)_{CO}$$

CI $\rightarrow$ cash inflow
CO $\rightarrow$ cash outflow
(-) because outflow is negative.

$$PW(10\%)_{CI} = (P/F, i, n)$$

$$P = F(1+i)^{-n} = 90,000(1.1)^{-1} + 80,000(1.1)^{-2} + 70,000(1.1)^{-3} + 60,000(1.1)^{-4} + 50,000(1.1)^{-5}$$

$$= 272,565.38 \quad 272,552.79$$

$$PW(10\%)_{CO} = -250,000$$

$$PW(10\%) = 272,565.38 - 250,000 = 22,565.38$$

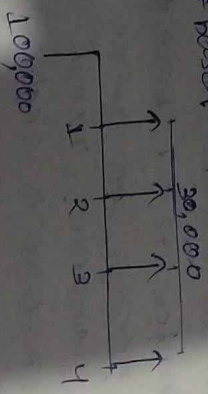
$$= 22,565.38 > 0, \text{ accept}$$

Q) project A cash investment proposal

end of yr	0	1	2	3	4
Net cash flow	-1Lakh	30,000	30,000	30,000	30,000

at $i = 10\%$ evaluate the project under PW method

net cash flows cash inflow cash outflow
* based on annuity $\rightarrow$ cash amount equal



$$PW(10\%) = PW(10\%)_{CI} - PW(10\%)_{CO}$$

$$PW(10\%)_{CI} = (P/A, i, n)$$

$$P = \frac{A}{i} \{1 - (1+i)^{-n}\}$$

$$A = 30,000$$

$$i = 0.10$$

$$n = 4 \text{ yrs}$$

$$P = 95,095.916 \sim 95096$$

$PV(10\%) = 95096 - 1,00000 = -49104 < 0$ (Reject)

Note
 In case of mutually exclusive project, calculate both A & B's present worth, the project whose present worth is higher or more in comparison that project is selected.

Future worth method is:

steps

- 1 Estimate the project length
- 2 calculate the rate of return
- 3 calculate the future worth of cash outflow
- 4 calculate the future worth of cash inflow
- 5 calculate the net future worth (NFW)
- 6 $NFW = FW(10\%)_{CI} - FW(10\%)_{CO}$
- 7 If $FW(10\%) > 0$, Accept
- 8 If $FW(10\%) < 0$, Reject

* when it is guarantee to get back the certain amount of money, or when we are getting guaranteed income from a project, this method is used.

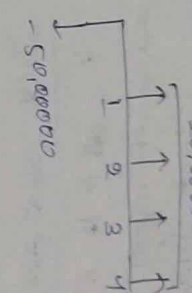
24 Nov 2020

Example

Rate of return (i), project length (n), CI, CO, FCI, FCI then $NFW(i) = FW(i)_{CI} - FW(i)_{CO}$

Given
 $i = 18\%$
 $n = 4 \text{ yrs}$

Project A



Project B

$FW = 18\% \text{ CI}$
 $FW = 18\% \text{ CO}$
 $FW = \frac{P}{i} [(1+i)^n - 1] - CI$
 $NFW(A) = 30,975,199$
 $CO = 50 \times 10^5 \times 1.18^4 = 96,93,888.799$

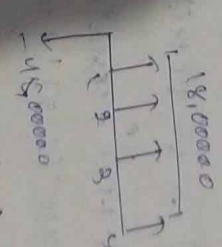
equal payment series based on annuity

$(F/A, i, n) - CI$
 $(F/P, i, n) - CO$

single payment compound interest
 compound interest
 compound amount factor

Project B

$i = 0.18$
 $n = 4 \text{ yrs}$
 $P = 40 \times 10^5$
 $A = 18 \times 10^5$



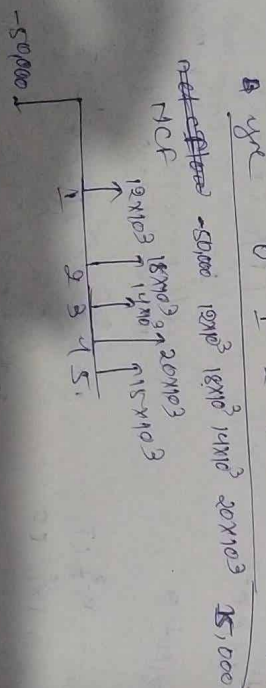
$FW(18\%) = \frac{A}{i} [(1+i)^n - 1]$
 $= 93,87,777.59 - CI$

$CO = 40 \times 10^5 (1+0.18)^4$
 $= 8,724,199.92$

$NFW(18\%) = 6,63,277.67$
 $NFW(A) > NFW(B)$

So, project A should be selected.

often the following cash flow information
evaluate its feasibility at
~~1-2-2~~ $i = 0.12$ under the method.



Net cash flow = $(CI - CO)$ net
based on single payment series
future worth.

$$FW(12\%) = FW(12\%)_{CI} - FW(12\%)_{CO}$$

$$(F/P, i, n) \rightarrow F = P(1+i)^n$$

$$= 12 \times 10^3 (1.12)^1 + 18 \times 10^3 (1.12)^2 + 14 \times 10^3 (1.12)^3 + 20 \times 10^3 (1.12)^4 + 15 \times 10^3 (1.12)^5$$

$$= 119,139.536$$

$$FW(12\%)_{CI} = 50,000 (1.12)^5$$

$$= 88,117.08 \sim 88,117$$

$$FW = 10,956$$

Internal rate of return

* each factor of production has rate of return.

Capital (if loaned) $\rightarrow$ interest

Labour $\rightarrow$ rent

At what rate of return we are going to compare,

$$\text{if } (PVI)_{CI} - (PVI)_{CO} = 0,$$

types (3 type)

① MARR $\neq$ minimum acceptable rate of return

* varies from organization to organization

on the basis of objective (depending upon their strategies & defn).

Ex: deposit $\rightarrow 10,000 \rightarrow 120,000 \rightarrow 1 \text{ yr}$, $10,000 \rightarrow$ monthly income.

min $\rightarrow 134,000$

1 year deposit/invest $\rightarrow$ end of each year 10%.

if MARR is more than 124,000 $\rightarrow$ less uncertainty, then we are going to invest.

② IRR: Internal rate of return

at what rate of interest present

worth of money $\rightarrow$ 20% cash inflow & cash outflow

$$PVI(1)_{CI} = PVI(1)_{CO}$$

or $PVI(1)_{CI} - PVI(1)_{CO} = 0$

suppose in bank as ~~20%~~ we get 4% interest

suppose we gave to a friend with

monthly (1) = 5%, & if we give that

to our brother, return = 0.

more rate of return $\rightarrow$ less present worth.

* External rate of return is Rate of return which is affected by the external environment changes going on in economic, political, social environment, external trade increase → more export of foreign currency

Foreign affairs department → fix external rate of return.

2 affecting international monetary fund policy (IMF)

It is 24 Nov. 2020

under IRR - 2 methods to used to determine IRR

- ① spread sheet method
- ② Trial & error method

Direct solution method.

ex:- given MARR = 20%

$yr(c_n) = 2\%$

$yr(c_n)$	0	1	2
net cash flow	-2000	1900	1500

Determine IRR & find its feasibility.

step

① $PW(i^*) = 0$

② $PW(i^*) = PW(i^*)_{01} - PW(i^*)_{10} = 0$

③ solve for i^*

④ $i^* = IRR$

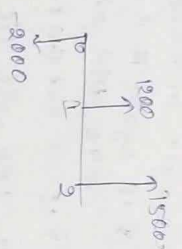
⑤ if $IRR > MARR$ accept

if $IRR < MARR$ reject

if $IRR = MARR$ - Remain indifferent

($i^* = IRR$).

solution



$$PW(i^*) = PW(i^*)_{01} - PW(i^*)_{10}$$

$$(P/F, i, n) \Rightarrow P = F(1+i)^{-n}$$

discounting at what rate of interest, the present value is 0.

$$1200(1+i)^{-1} + 1500(1+i)^{-2} - 2000 = 0$$

* present worth of cash outflow always

remains same.

$$\frac{1200}{(1+i)} + \frac{1500}{(1+i)^2} - 2000 = 0$$

$$1200a + 1500a^2 - 2000 = 0$$

$$1200a + 1500a^2 - 2000 = 0$$

$$1200a + 1500a^2 - 2000 = 0$$

$$1200a + 1500a^2 - 2000 = 0$$

$$1200a + 1500a^2 - 2000 = 0$$

$$1200a + 1500a^2 - 2000 = 0$$

$$10 = 8 + 8i^*$$

$$i^* = \frac{2}{8} = 25\%$$

$$IRR = 25\%$$

As $IRR > MARR$, accept the project.

Trial and Error method

also known as hit and trial method.

suppose $MARR = 10\%$

let $IRR = 8\%$

- ① (1st take $IRR < MARR$)
- ② find present worth
- ③ if present worth is (+)

again increase it to 12%

③ If again it is (+) PV, then take IRR at 14%

④ If we get (+) PV at 14%, so, IRR will lie in between (12% - 14%)

new value of IRR is used the method of Interpolation

mean mathematical average

median, mode - positional average

↓

frequency repeated max then

1/2 of graph

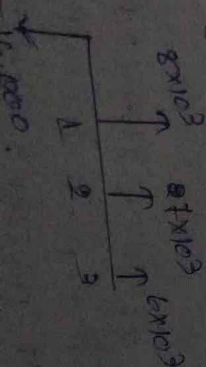
moving average - suppose there are 5 variable (a,b,c,d,e) have to calculate 3yr moving avg, then $\frac{a+b+c}{3}$, $\frac{b+c+d}{3}$, $\frac{c+d+e}{3}$

* ~~mean~~ & mode & median are based on interpolation.

$$\text{Net value of IRR} = 12\% + \frac{PV(12\%) - PV(14\%)}{PV(12\%) - PV(14\%)} \times 2\%$$

Ex 7.3 yr.

CF	8000	7000	6000
yr	0	1	2



MARR = 12%

Let $i = 10\%$

$$PV(10\%) = PV(10\%) - PV(10\%) = 0$$

$$P = F(1+i)^{-n}$$

$$PV(10\%) = 8000(1.1)^{-1} + 7000(1.1)^{-2} + 6000(1.1)^{-3} - 16000$$

$$= 1565.74$$

Let $i = 14\%$

$$PV(14\%) = 8000(1.14)^{-1} + 7000(1.14)^{-2} + 6000(1.14)^{-3} - 16000 = 453.64$$

Let $i = 16\%$

$$PV(16\%) = 8000(1.16)^{-1} + 7000(1.16)^{-2} + 6000(1.16)^{-3} - 16000 = -57.36$$

so, IRR lies in between 14 to 16%

new value of IRR

$$= 14\% + \frac{453.64}{453.64 - (-57.36)} \times 2\%$$

$$= 15.77\% \sim 15.8\%$$

MARR given = 12%

Since $IRR > MARR$, so the project should be accepted.

should be accepted.

Mutually exclusive project:-

The happening of one let other not

to happen.

It is based on probability & helps in formulating theoretical distribution.

95% confidence level

5%
99%

at: after 2020

The method of accept or reject based on IRR is not applicable here.

An mutually exclusive project following procedure steps are considered.

Find out which cash flow investment is lower & which one is higher.

Suppose project A cashflow $\rightarrow$ lower
B $\rightarrow$ higher.

Subtract the cash flow of lower investment i.e. project B from higher investment i.e. project A

The incremental cash flow is the difference between the IRR on these incremental cash flow.

Determine the resultant IRR i.e., $IRR(B-A)$. The decision rule is

If $IRR(B-A) > MARR$ select A
If $IRR(B-A) < MARR$ select B
If $IRR(B-A) = MARR$ either one

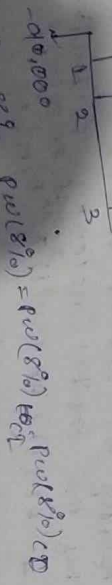
of end year	Project A	Project B
0	-30,000	-1,20,000
1	13,500	42,000
2	18,000	62,250
3	15,000	63,300

MARR = 10%

Soln incremental cash flow b/w 2 projects

end year	incremental cash flow (B-A)
0	-1,20,000 - (-30,000) = -90,000
1	+12,000 - 13,500 = +28,500
2	+18,000 - 18,000 = 0
3	+15,000 - 15,000 = 0

incremental cash flow diagram



$$let\ I = 8\%$$

$$= P(1, I, n) = P \times \frac{1 - (1 + I)^{-n}}{I}$$

$$PW(8\%) = 28,500(1.08)^{-1} + 0 + 0 = 26,500(1.08)^{-2} + 0 = 21,602.31$$

$$PW(14\%) = 28,500(1.14)^{-1} + 0 + 0 = 25,000(1.14)^{-2} + 0 = 19,662$$

$$PW(16\%) = 28,500(1.16)^{-1} + 0 + 0 = 24,500(1.16)^{-2} + 0 = 18,662$$

$$PW(18\%) = 28,500(1.18)^{-1} + 0 + 0 = 23,500(1.18)^{-2} + 0 = 17,662$$

$$PW(20\%) = 28,500(1.20)^{-1} + 0 + 0 = 22,500(1.20)^{-2} + 0 = 16,662$$

$$PW(22\%) = 28,500(1.22)^{-1} + 0 + 0 = 21,500(1.22)^{-2} + 0 = 15,662$$

$$PW(24\%) = 28,500(1.24)^{-1} + 0 + 0 = 20,500(1.24)^{-2} + 0 = 14,662$$

$$PW(26\%) = 28,500(1.26)^{-1} + 0 + 0 = 19,500(1.26)^{-2} + 0 = 13,662$$

$$PW(28\%) = 28,500(1.28)^{-1} + 0 + 0 = 18,500(1.28)^{-2} + 0 = 12,662$$

$$PW(30\%) = 28,500(1.30)^{-1} + 0 + 0 = 17,500(1.30)^{-2} + 0 = 11,662$$

$$PW(32\%) = 28,500(1.32)^{-1} + 0 + 0 = 16,500(1.32)^{-2} + 0 = 10,662$$

$$PW(34\%) = 28,500(1.34)^{-1} + 0 + 0 = 15,500(1.34)^{-2} + 0 = 9,662$$

$$PW(36\%) = 28,500(1.36)^{-1} + 0 + 0 = 14,500(1.36)^{-2} + 0 = 8,662$$

$$PW(38\%) = 28,500(1.38)^{-1} + 0 + 0 = 13,500(1.38)^{-2} + 0 = 7,662$$

$$PW(40\%) = 28,500(1.40)^{-1} + 0 + 0 = 12,500(1.40)^{-2} + 0 = 6,662$$

$$PW(42\%) = 28,500(1.42)^{-1} + 0 + 0 = 11,500(1.42)^{-2} + 0 = 5,662$$

$$PW(44\%) = 28,500(1.44)^{-1} + 0 + 0 = 10,500(1.44)^{-2} + 0 = 4,662$$

$$PW(46\%) = 28,500(1.46)^{-1} + 0 + 0 = 9,500(1.46)^{-2} + 0 = 3,662$$

$$PW(48\%) = 28,500(1.48)^{-1} + 0 + 0 = 8,500(1.48)^{-2} + 0 = 2,662$$

$$PW(50\%) = 28,500(1.50)^{-1} + 0 + 0 = 7,500(1.50)^{-2} + 0 = 1,662$$

$$PW(52\%) = 28,500(1.52)^{-1} + 0 + 0 = 6,500(1.52)^{-2} + 0 = 0,662$$

$$PW(54\%) = 28,500(1.54)^{-1} + 0 + 0 = 5,500(1.54)^{-2} + 0 = -0,338$$

$$PW(56\%) = 28,500(1.56)^{-1} + 0 + 0 = 4,500(1.56)^{-2} + 0 = -1,338$$

$$PW(58\%) = 28,500(1.58)^{-1} + 0 + 0 = 3,500(1.58)^{-2} + 0 = -2,338$$

$$PW(60\%) = 28,500(1.60)^{-1} + 0 + 0 = 2,500(1.60)^{-2} + 0 = -3,338$$

$$PW(62\%) = 28,500(1.62)^{-1} + 0 + 0 = 1,500(1.62)^{-2} + 0 = -4,338$$

$$PW(64\%) = 28,500(1.64)^{-1} + 0 + 0 = 0,500(1.64)^{-2} + 0 = -5,338$$

$$PW(66\%) = 28,500(1.66)^{-1} + 0 + 0 = -0,500(1.66)^{-2} + 0 = -6,338$$

$$PW(68\%) = 28,500(1.68)^{-1} + 0 + 0 = -1,500(1.68)^{-2} + 0 = -7,338$$

$$PW(70\%) = 28,500(1.70)^{-1} + 0 + 0 = -2,500(1.70)^{-2} + 0 = -8,338$$

$$PW(72\%) = 28,500(1.72)^{-1} + 0 + 0 = -3,500(1.72)^{-2} + 0 = -9,338$$

$$PW(74\%) = 28,500(1.74)^{-1} + 0 + 0 = -4,500(1.74)^{-2} + 0 = -10,338$$

$$PW(76\%) = 28,500(1.76)^{-1} + 0 + 0 = -5,500(1.76)^{-2} + 0 = -11,338$$

$$PW(78\%) = 28,500(1.78)^{-1} + 0 + 0 = -6,500(1.78)^{-2} + 0 = -12,338$$

$$PW(80\%) = 28,500(1.80)^{-1} + 0 + 0 = -7,500(1.80)^{-2} + 0 = -13,338$$

$$PW(82\%) = 28,500(1.82)^{-1} + 0 + 0 = -8,500(1.82)^{-2} + 0 = -14,338$$

$$PW(84\%) = 28,500(1.84)^{-1} + 0 + 0 = -9,500(1.84)^{-2} + 0 = -15,338$$

$$PW(86\%) = 28,500(1.86)^{-1} + 0 + 0 = -10,500(1.86)^{-2} + 0 = -16,338$$

$$PW(88\%) = 28,500(1.88)^{-1} + 0 + 0 = -11,500(1.88)^{-2} + 0 = -17,338$$

$$PW(90\%) = 28,500(1.90)^{-1} + 0 + 0 = -12,500(1.90)^{-2} + 0 = -18,338$$

Annual Equivalent Method

* This method is used to calculate both single as well as mutually exclusive investment proposals, applicable to evaluate under this method an equivalent annual payment made on an annual basis is to be calculated.

This method consists of following steps
 ① compute the net present worth of the cash flow stream.

② multiply this amount of present worth by the capital recovery factor, i.e.,

$$AEC(i) = PW(i) \times \frac{i(1+i)^n}{1-(1+i)^{-n}}$$

③ the decision rule for single project is

If $AEC(i) > 0$ accept

If $AEC(i) < 0$ reject

$AEC(i) = 0$ remain indifferent

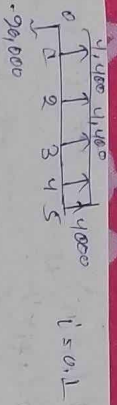
④ out arise of mutually exclusive project the min annual equivalent cost will be selected.

or the alternative with the min annual equivalent ~~cost~~ will be selected as the best alternative.

Ex: - 10/12/2020

Q) cost of machine 20,000, estimated life 5 yrs. At the end of 5 yrs it can be sold for Rs 4000 after an adjustment. if the firm could earn Rs 4,400 per year, with this machine, should it be purchased at the interest rate of 10%?

Sol



$$C_0 = -20,000$$

Scrap value

$$4000 = \text{Scrap value}$$

$$PW(10\%) = PW(10\%)_{CF} + PW(10\%)_{SV}$$

$$A = 4,400$$

$$= \frac{A}{i} \left[1 - (1+i)^{-n} \right] + F \left[(1+i)^{-n} \right]$$

$$= \frac{4,400}{0.10} \left[1 - (1.1)^{-5} \right] + 4000(1.1)^{-5} = 20,000$$

$$= -236.85$$

$$AEC(10\%) = PW(10\%) \times \frac{i(1+i)^n}{1-(1+i)^{-n}}$$

$$= -236.85 \times \frac{0.1(1.1)^5}{1-(1.1)^{-5}}$$

$$= -220.45$$

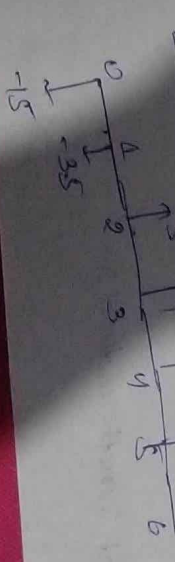
Capital Recovery factor = $\frac{i}{1-(1+i)^{-n}}$

we may reject

since $AEC(10\%) < 0$, the project as machine does not generate sufficient revenue to recover the original investment

Example 4.5 (Annual eq. method)

Year	0	1	2	3	4	5	6
Cash flow	-15	-3.5	5	9	12	10	8



$$\begin{aligned}
 PW &= (P/A, i, n) \quad i = 15\% = 0.15 \\
 &= F(1+i)^{-n} \\
 &= 5(1.15)^{-2} + 9(1.15)^{-3} + 12(1.15)^{-4} + \\
 &\quad 10(1.15)^{-5} + 8(1.15)^{-6} + 3.5(1.15)^{-7} \\
 &= 21.946
 \end{aligned}$$

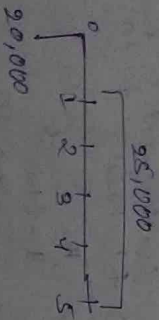
$PW > 0$ accept project.

$$PW(18\%) = 24.62 - 15.69 = 6.946$$

$$9.5 = \frac{14.62}{0.15} - 6 = 1.83$$

accept project.

$i = 12\%$ (Eq present worth method)



$$P = 25,000$$

$$n = 5 \text{ yrs}$$

$$PW(12\%) = PW(12\%)_{CF} - PW(12\%)_{CO}$$

$$= \frac{25,000}{0.12} \left[1 - (1.12)^{-5} \right] - 20,000$$

$$= 40,119.4$$

cash outflow = 25,000,000.

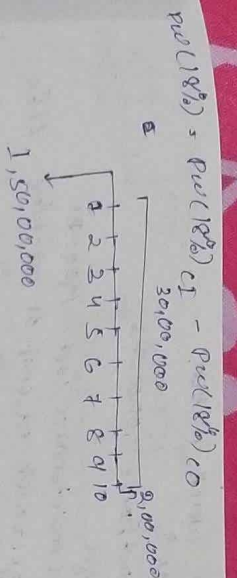
annuity/annual equivalent yield = 30,00,000

$$n = 10 \text{ yrs}$$

salvage value after 10 yrs = 2,00,000

$$i = 18\%$$

Eq present worth method is



$$PW(18\%) = PW(18\%)_{CF} - PW(18\%)_{CO}$$

$$= \frac{30 \times 10^5}{0.18} \left[1 - (1.18)^{-10} \right] + 2 \times 10^5 (1.18)^{-10}$$

$$= -1,479,528.22$$

since $PW(18\%) < 0$, the rejection of oil well is not financially feasible.

Eq

present worth method:

$$i = 10\%$$

cash outflow = 5,00,00,000

$$n = 10 \text{ yrs}$$

$$P = 50,00,000$$

modernising cost at end

of 10 yrs = 2,00,00,000

salvage value at end of 10 yrs = 1,00,00,000

$$PW(10\%) = PW(10\%)_{CF} - PW(10\%)_{CO}$$

$$= \frac{50,000}{0.10} \left[1 - (1.1)^{-10} \right] + 2 \times 10^5 \times (1.1)^{-10} +$$

$$- 1 \times 10^5 \times (1.1)^{-10} - 5,00,000$$

since PW is less than 0, the project is not feasible.

10) future worth method

$i = 15\%$

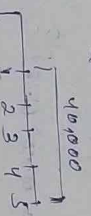
cash outflow = 4,00,000 RS

$n = 5$ yrs

annual maintenance cost = 40,000 RS

salvage value at end of machine life

= 2,00,000 RS



$$FW(15\%) = FV(15\%)_{C_1} - FV(15\%)_{C_2}$$

$$= \frac{40,000}{0.15} [(1.15)^5 - 1] + 2,00,000 \times (1.15)^0$$

$$= [4,00,000 \times (1.15)^5 - 2,00,000] = -334,844.44$$

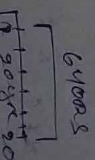
not feasible $\rightarrow$ cash outflow

uniform annual benefit (A) = 64,000 RS

$n = 20$ yrs.

$i = 15\%$

future worth method



$$FV(15\%) = FV(15\%)_{C_1} - FV(15\%)_{C_2}$$

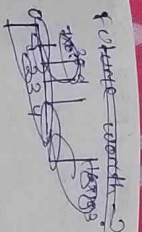
$$= \frac{64,000}{0.15} [(1.15)^{20} - 1] - [4,00,000 \times (1.15)^0]$$

$$= 97,7433$$

New 10, Hence the project is economically desirable.

12) $i = 15\%$

end of yrs	0	1	2	3	4	...	10
cash flow (RS)	-	7,000	8,000	9,000	10,000	...	16,000



future worth $\rightarrow$

$$FV(15\%) = FV(15\%)_{C_1} - FV(15\%)_{C_2}$$

$$= 7,000 \times (1.15)^9 + 8,000 \times (1.15)^8 + 9,000 \times (1.15)^7 + 10,000 \times (1.15)^6 + 11,000 \times (1.15)^5 + 12,000 \times (1.15)^4 + 13,000 \times (1.15)^3 + 14,000 \times (1.15)^2 + 15,000 \times (1.15)^1 + 16,000 \times (1.15)^0$$

$$= 210,817.48$$

14-15/12/20

Cost Benefit Analysis

* Benefit cost analysis is a technique used to evaluate public as well as private investment project.

* In private investment before investing individual have to consult & evaluate it properly.

* Public project $\rightarrow$ project undertaken by govt, municipalities, panchayat, local authorities.

note $\rightarrow$ provide "net" benefit to public they have to evaluate it & find out the best alternatives to give

Three activity undertaken by govt. (Public activity)

- ① protect & culture development
- ② employment benefit.
- ③ Economic benefit.

→ protection & military: protection from hazards
 → employment & culture development: -
 public school system, library, post, recreation, facilities & knowledge to people

recreation facilities (museum, park etc)

→ research & development

→ economic benefit & for increasing trade policy, internal & foreign trade govt of india improve working way, food control, research & development policies.

* govt of india has to ^{use} ~~the~~ some techniques which will help to evaluate the individual project & spell out whether it will uplift the national economy with limited resources available.

benefit cost analysis (BCA)
 methodology developed to evaluate all type of project with social point of view

by taking into account BCI

- ① ~~investment~~ cost
- ② operation & maintenance
- ③ ~~investment~~ cost

↳ program / innovation

(B/I etc)

method used.

min^m cost & max^m benefit → main theme.

BCI = eq. benefit
 eq. cost

Eq. benefit

the benefit the gaining into consideration a particular time factor (P, F, W, D, W)
 Equivalent cost
 total resources we have to utilize with making the project available for public.
 at a particular time factor.

BCI > 1 - accept.

BCI < 1 - reject

BCI = 1 depends on govt whether to accept or reject.

single investment

~~cost analysis~~ ^{benefit} CB / BCI is used to evaluate

* methodology which is used from social all the investment projects from social point of view.

* It spell out national economic objectives & budget allocation of resources, to various sectors.

* CBA can be used to evaluate single public investment project as well as mutually exclusive alternatives.

procedure of CBA to evaluate single public projects:-

- ① Forecast the cost & benefits attached to the proposed project.
- ② Determining the proper discount rate.
- ③ convert the benefits (income) and cost

into present value. i.e.,

Find out the benefit to cost ratio

$$BC(i) = \frac{\text{Eq. benefit}}{\text{equivalent cost}}$$

or

$$\text{present worth of}$$

$$BC(i) = \frac{B_p}{i + C_p}$$

BC benefit ~~cost~~ is initial investment

BC present worth of benefit

initial investment = present worth of investment

CP present value of cost

annual operation & maintenance cost

accept and decision $\rightarrow$ $BC(i) > 1$ accept

BC(i) = $\frac{B}{i + C}$ < 1 Reject

BC(i) = $\frac{B}{i + C} = 1$, decision depends on

Example given

I = 50,00,000

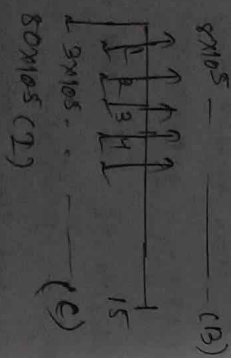
n = 15 yrs

C = 3,00,000

B = 8,00,000 / yr.

discount rate = 10% = i

$\downarrow$ cash outflow



$$BC(10\%) = \frac{B_p}{1 + C_p}$$

$$BC = \frac{8 \times 10^5}{0.10} \{ 1 - (1.1)^{-15} \}$$

$$= \frac{8,00,000}{0.10} \{ 1 - (1.1)^{-15} \} = 6,084,863.60$$

$$BCP = \frac{3 \times 10^5}{0.1} \{ 1 - (1.1)^{-15} \}$$

$$= \frac{3,00,000}{0.1} \{ 1 - (1.1)^{-15} \} = 2,281,829.85$$

$$1 + C_p = 7,281,829.85$$

$$BC(10\%) = \frac{BCP}{1 + C_p} = \frac{2,281,829.85}{7,281,829.85} < 1$$

since $BC(10\%) < 1$, the construction of dam is not justified.

At 10% discount rate

Benefit cost ratio to evaluate mutually exclusive project:-

In case of mutually exclusive project, the benefit cost ratio is calculated on incremental basis.

procedure for evaluation of mutually exclusive projects:-

1) Find out the benefit cost ratio of all projects available

2) eliminate the projects which have value as per benefit to cost ratio.

3) The remaining alternatives which have benefit cost ratio greater than 1 (> 1)

should be arranged in increasing order of the cost (C).

④ The alternative which has the lowest cost should have the 1st preference. The second smallest & so on.

⑤ Calculate the incremental differences of benefit and cost for the paired alternatives.

⑥ Compute the benefit to cost ratio on incremental investment by following the formula $BC(i) = \frac{AB}{AI+AC}$

where AB = difference b/w the benefit of two pairs project X and Y

AI = diff. b/w the initial cost of project X & project Y.

AC = difference b/w the ~~cost~~ annual operation & maintenance cost of project X & Y.

If $\frac{AB}{AI+AC} > 1$, prefer ^{project} X over Y.

If $\frac{AB}{AI+AC} < 1$, prefer project Y over project X.

⑦ Compare the other projects to find out the best one.

Ex: $i = 10\%$

Component	Project X	Project Y	Project Z
I	50,000	8,00,000	1,40,000
B	1,20,000	3,50,000	9,10,000
C	40,000	80,000	10,000
$PW(i)$	30,000	70,000	60,000

* $B - (I+C)$ (present worth) = $PW(i)$

⑧ If all 3 projects are independent & $PW(i)$ is positive for all the 3 projects, so, all projects should be acceptable, if they were independent.

⑨ If these three projects are mutually exclusive,

$$PW(10\%) X = \frac{B}{I+C} = \frac{1,20,000}{40,000} = 1.33$$

$$PW(10\%) Y = \frac{B}{I+C} = \frac{3,50,000}{9,10,000} = 1.25$$

$$PW(10\%) Z = \frac{B}{I+C} = \frac{9,10,000}{1,50,000} = 1.40$$

As $B(10\%) > 1$, so all projects are justifiable.

⑩ When all the projects are mutually exclusive, in this case, we must use the pairwise of incremental analysis, i.e., by arranging the projects by increasing order of their denominator i.e., $(I+C)$

Ranking	X	Z	Y
$I+C$	40,000	1,50,000	9,10,000

Ranking $X \rightarrow Z \rightarrow Y$
1st compare X & Z, then the one better b/w X & Z will be compared with Y.

$$\frac{PW(10\%) Z - X}{B - (I+C)} = \frac{B}{AI+AC} = \frac{9,10,000 - 1,50,000}{(1,50,000 - 40,000) + (9,10,000 - 1,50,000)}$$

8

$$= \frac{90,000}{60,000} = 1.5 > 1$$

So, 2 is preferable than 1.
comparison 1 & 2 or (1 vs 2)

$$OC(10\%)_{1-2} = \frac{10}{11+12}$$

$$= \frac{3,50,000 - 2,10,000}{1,30,000}$$

$$= \frac{1,40,000}{1,30,000} = 1.076 > 1$$

$$(2,00,000 - 1,40,000) + (1,80,000 - 1,00,000)$$

$$= \frac{1,40,000}{1,30,000} = 1.076 > 1$$

So, we prefer 2 over 1.

So, we can conclude 2 is the ultimate choice.

Advantage of Benefit cost method

- * This method helps us to eliminate the inferior & useless project.

- * Based on the benefit & cost the project is selected/approved, for inferior project there is no political pressure & public bodies taken into consideration.

- * It takes into consideration / convert the qualitative data into quantitative form.

- * For mutually exclusive project that is the most acceptable method of evaluating public project

~~* Benefit cost ratio~~

Limitation of Benefit to cost ratio

It is based on assumption which is unrealistic in nature.

assumptions

① there is full employment in economy

② the economy is a market economy (agriculture is only market economy)

③ (an India) (agriculture is more)

④ we are ignoring the opportunity cost, (e.g. unable to examine financial aspect)

⑤ It only tells about financial aspect not technological aspect.

⑥ It has been illegal (because we have control over public project is (Real capitalism) → pen & paper project is there only,

there only,

Banking

The term bank is derived from ^{an} banker which means a bank note from which the European money lender used to exit money

- * It is an financial institution which grants loans & accept deposits from public.

- * A bank being a buyer & seller of money stimulates economic activity by mobilizing the ^{saver} money of people

- * If the banking system of the country is developed, then economy

→ growing very fast, establishment expenditure of bank is going to be enriched.

* In 1914, Indian Banking Regulation Act means accepting money from public for the purpose of lending or investing. These deposit may be money payable on by cheques, bill of exchange.

* Bank is an financial institution who mobilises surplus funds by lending to public / needy businessmen / organisation by way of interest.

* main motto of profit making by lending money to individuals & Bank is.

* Commercial bank - main objective is to earn profit by granting loan with higher rate of interest & accepting money with lower rate of interest.

* It is an institution which is ~~used~~ to arrange, to use the surplus fund of public by making it available to those who need it.

* Bank is a middleman agent who mobilises the surplus money of public by granting loan to trade, industrial & commerce sectors, to

act as ^{settling} their revenue & capital expenditure. ~~acceptor~~ & debtor. It is an entity that extends credit, giving another entity permission to borrow money to be repaid in the future. ~~debtor~~ the person from who business will receive the benefit given earlier.

* creation of money when the depositors going to withdraw the extra amount of money than that

is existing in a account & called ~~existing~~ customer's facility as next bank provides better facility as next basis.

* Instrument of innovation carrying out all type of agency function. collect interest, offer service.

* It has become a part of society & ways to make in civilisation.

diff type of bank

In yr 1947 nationalisation of bank started in phases:-
① 1955 - state became state

② 1961 - 14 major banks
③ 1980 - Andhra Bank, Punjab

In yr 2000, we have only 12 nationalised bank
primary form of commercial

* accepting deposit from public

- ① fixed deposit
- ② current deposit
- ③ saving deposit
- ④ recurring deposit

if for fixed period it is deposited then earn at a higher interest.

② savings, no days, etc...
③ facility given to business man, no interest is allowed by bank as there is no reservation in withdrawal.

④ min 500 deposit, with cheque facility 1000 - cashless, now zero balance

no restriction on no. of times of withdrawal.

4) on order these accounts.

bank opens a fixed amount at the parents deposit, at the end of period, at the end of every year, at the end of period, the interest is accumulated & we get the maturity amount $\rightarrow$ R.D. money + interest.

when we are unable to pay the bank is going to change some penalty

* providing bank to needy borrowers.
① money at all $\rightarrow$ given from short period (1-14 days)

provided to one bank from other bank, when there are in urgent need, need not be in need of high rate of interest, not given to public.

② overdraft need to be ordinary people by keeping some ornaments in return get loan

bank is credit creator $\rightarrow$ when the loan is, create an account in bank.

suppose loan amount - 1 lakh, 30000 have taken, 70000 remaining, going to use by bank to grant loan to other persons. it helps to generate more with interest.

③ cash credit
④ over draft $\frac{DT + 31/12/20}{\text{discarding of bill of exchange}}$

in 1818 negotiable instrument act has been established.

① bill of exchange
② promissory note
③ cheque

* There are some other document through which bill of exchange can be negotiated.

which bill of exchange is negotiable document containing a certain amount of money on a particular time later, maturity time period.

* 30 days of grace to be added
Get approval if written then $1 + \frac{3}{100}$ $\times$ actual date.

* 30 days of grace

* Before 2 month we can discount the bill
suppose bill amount - 8000, 6% commission the bank will charge
annum, 480 $\rightarrow$ 5000 commission.

* discount $\rightarrow$ loss to the drawer.
 $\rightarrow$ it helps the business man to discount the bill before time period.

① short term - 1 year

②

③ long term $\rightarrow$ more than 1 year $\rightarrow$ 300 months.

In case of term loan, the B rate of interest is more monthly payment.

* out type of consumable goods can be bought by help of bank

* ~~Bank~~ investment of funds.

the funds are invested in highly liquid asset i.e. these assets which can be ~~converted~~ converted into cash

* immediately loan & advancing i.e. when bank generating loan & advancing people to open account & loan, the bank ensures a deposit of credit, the loan created is alone. thus more so credit creation is alone. thus more interest can be earned.

* Advancing the use of cheque i.e. ~~that~~ it is a written document issued by bank where we can put the name of person & amount of money which will be deducted from our account & credited to that person's account.

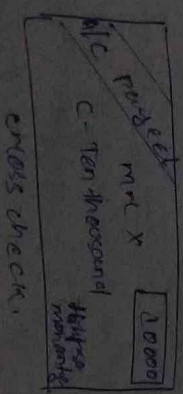
off type

① open check :- There is no safety & security

② crossed check :- i must have account, the bank is not going to make transaction of cheque at cash counter.

cheques

* most advanced instrument



open check
 1. bearer check
 2. the person who can cash the check
 3. order → the person ordered by can cash the money

the person who can cash the check can cash the money

* the bank also provide gift check.

agency function

* with a nominal cost bank provide some services to its customer → agency function

* national electronic fund transfer (NEFT)

Real time gross settlement (RTGS)

② Bank draft → the written document issued by another bank on one bank.

* NEFT - any amount

* RTGS - 2 lakh or more

RTGS → Real time - no waiting period
 Gross settle → transaction is made on one basis including total amount etc.

NEFT

on batches

RTGS
 individually, one after another

③

mobile, gas, monthly magazine subscription

④ also purchase security on behalf of customer

⑤ Act as a trustee and executive, the suppose a person's will is there, the

bank is going to exercise the call:

- act as counterpart / agent or representative
- active or passive, are picked up or obtain passport, are picked up
- reference in

⑦ helps the customer is
conscientiously is free

General utility function

Marker facility
at the end of each yr.
renewal must be
renewed before 30 Nov.

③ prenatal / tissue precursors change on request
so that they can travel without fear
of loss of memory

③ letter of credit is the bank act as a guarantied, it is alone to satisfy credit

② " letter of credit is mainly used for foreign trade.

⑦ collection of vital statistics - 2 of foreign
about commerce, banking, investment
thru agents & passports. foreign exchange
⑧ understanding of shares & adventure.

private company more than
50% held by govt
→ controlled is thors

* ~~2006~~ 2006 member
* put Unlabeled at end.

*agrowth company can issue share & openenture

* eleventh
and then placement which acknowledged
the debt.

* share
when public is going to buy

* 10% subscription should be made by
all the shareholders.

* benefit as a multiphase man & trained nurse
to subscribe these changes by public &
also change commission.

credit card facility
exchange business

Foreign ex.

5 January 2021

Reserve Bank of India,

The need of Raj 1973, when
warren had been, governed of
Bangalore proposed the establishment of a
central bank in Bangalore & Bihar
passed Sept. 8- Bill was passed
under Bank Act) 1905

assembly (assembly) 1. Fundraising April 1955
 It started with 1 fundraising bank
 It was constituted as a shareholder bank
 formed by capital 5 units, 2.2 March
 subscribed by central gov't, west public (local)
 subscribed by central gov't, west public (local)

the decision re
1914, January
1919, ~~1920~~, 1921
auto record.

* started functioning as an
initiation, compensation - 118.10rs to
grainate stage before [REDACTED] consitituted

* management or
govt members:-
but ~~the~~ governed.

4 - deputy director of environment
4 - director, 1 from each board

to directors
1 govt of India } nominated by central govt.

functions of RBI ^{issuance}

* main funⁿ is to regulate the bank notes & to keep reserve w/ the money supply & to keep reserve w/ the money supply.

* control the amount & supply of currency.
* control the Banking Regulation act, money power to control the commercial bank.

① * monopoly of note issue (except IRS notes & coins)
↳ issue → holds asset against which currency can be issued.
↳ circulation & withdrawal of money is regulated by.

* assets consists of gold coins, foreign security bill of exchange & promissory notes.
there must be a proportional Reserve system, 10% must be coins.

③ transfer banking business from ~~govt~~ to section 20, 21, 21A etc.

↳ transfer to govt.

the bank also entered into separate agreement with govt before any transaction, for that no remuneration is provided to bank.

why carrying out this particular function

⑤ otherwise the bank to make way & means of advance to central & state govt.

⑥ Economic condⁿs are not stable, while formulating policy, RBI gives advice to

central govt. } central matters
* it includes banking & financial matters, consents in agricultural credits, (advancing to govt.)

④ control of credit ^{responsible for money}

the bank is responsible for money supply & to usual instrument "central bank" credit control

3 steps

① bank rate

↳ section 49 of act defines as stipulated rate the bank is going to purchase bill of exchange from outsiders (promissory notes) rate called as it controls the other interest rates of markets.

② Open market operation

the central bank is going to purchase govt & call securities, when there is going to surplus money in market it is going to sell securities & when there is shortage to sell securities & purchase credits. need of money, it purchase credits.

Security it is a borrowed money & sure to get it back at any time.

long term → to provide wide & healthy market for govt. security, short term to assist monetary security, motto → in terms of public interest.

fixed capital → the working capital → necessary for day to day life.

③ control of credit

* RBF is empowered to control the advance of Bank.

① Method of setting credit limit margin for lending against specific security.

② setting the amount of credit for certain purposes.

③ setting & fixing:

④ discriminatory rate of interest for certain deposits / advances.

* The overall growth of nation depends of Regional balance, which is maintained by discriminatory rate of interest.

* Statutory liquidity Ratio Requirement, CLR & statutory liquidity Ratio

* The commercial bank offset the impact of variable reserve ratio by liquidating the govt. securities.

- To prevent this action, pt of the Banking Regulation Act prescribed that Bank have to maintain liquid asset in cash, gold & securities not less than 20% of total demand & time liabilities.

In Apr 1962, ~~every bank has to~~ the act was amended that every bank should maintain min amount of liquid assets equal to 28% of the demand & time liabilities.

Liability asset

those assets which can be converted into cash immediately.

Cash Reserve Ratio

* The scheduled commercial banks are required to maintain a min cash balance with the Reserve bank by varying the Reserve & bank could control over the Reserve

Pl. 12 Jan 2021

* Finance is the ~~the~~ livelihood of every economy. In order to attain the welfare of people, we need to produce & distribute goods & services.

* The activities of producing & distributing goods & services comes under Real sector of economy.

* The fund required for this must be timely made, i.e., price, time, quantity ~~activity~~ needs to be considered.

* Financial system - consists the arrangement of devising ways in which we can mobilize the real sector of economy

operation & growth
① * Financial system, market, financial institution, constituents of financial system.

~~agreement~~ * structure & aspect of the interest rate system
② * structure & aspect of the interest rate system
③ * interrelationship of financial institution, how to mobilise financial institution to get more profit (money)

System

arrangement of elements together that develops a well that help us to facilitate growth & operation of economy

② Role played by system → analyse the role of financial system whether it suits & give objective satisfaction.
Role of financial system

* service render / credit facility, provide that how & can take financial help from economy / provision of raising funds

constituent of financial system

* Financial institution & banking & non-banking sector, able to mobilise the excess funds of people, mobilise saving

non-banking sector →

LIC, GIC, mutual invest, provident fund, unit trust, provident fund, etc. / R.F. → income tax

recognise R.F. → income tax
statutory R.F.
public R.F.

R. → employers contribute over 10% of employee salary → transfer

U → not recognised by income tax department.
not recognised by income tax department.

P → in each nationalised bank 5% is available
min 8000 max 70000 per month.
must be withdrawable by 50 Rs.

with the help of this fund the govt. utilise this money for public welfare.
general insurance & property marine insurance & any calamities / import, export.

1993, 1994, → intermediate → provide by rules & regulation of RBI & also provide fund / money to people.

* Financial market → exchange / ^{of} financial instruments are going on that place.

Financial instruments are available. money market → market for short term funds. (less than 1 year)

market for long term funds

money market provides a channel for exchange of financial funds

for money.

→ controlled by central bank (RBI)

→ working capital (provided by money market)

acquired to maintain & manage the day to day established expenses.

offer the commercial bank &

central bank acts as a developer & promoter in the financial market.

J.S. 10

Money market

Money market means the centre in which financial institutions converge for the purpose of that are closest substitutes of money, dealing in monetary asset

Characteristics of money market

- arrangement of short term funds
- tool for preparation of monetary policy & financial management
- it deals in high liquid instruments
- billions → ~~hundreds~~ ^{trillions} of rupees
- they have prime role kept in form of treasury / govt bond.

eg: govt (high liquid instruments)

- main role is played by RBI
- the players in this market are RBI & commercial bank
- consistent → commercial bank
- subject to RBI regulation

6 question

① MCQs 10x1 = 10 marks - compulsory

② SAQs 5x2 = 10 marks

selective

- ③ 8
- ④ 8
- ⑤ 8
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→ Depreciation theory & fixed installment & other methods

mutual difference, factors affecting depreciation.

Evaluation of engineering project

- own satisfy our own need. private activity
- passed by govt. for betterment of public.
- public activity.
- cost → sacrifice we have to made to acquire the resource.

Net present worth method

- project length (n)
- rate of return (i)
- factor of production - material, labour, capital, management

→ cash inflow (benefit)

→ cash outflow (cost)

→ net cash flow (benefit-cost) (NPW)

→ present worth of net cash flow (NPW)

if NPW > 0 accept

if NPW = 0 indifferent

if NPW < 0 reject

if NPW is negative indifferent

if NPW is zero indifferent

if NPW is positive accept

if NPW is negative reject

if NPW is zero indifferent

if NPW is positive accept

if NPW is negative reject

if NPW is zero indifferent

if NPW is positive accept

if NPW is negative reject

if NPW is zero indifferent

if NPW is positive accept

if NPW is negative reject

if NPW is zero indifferent